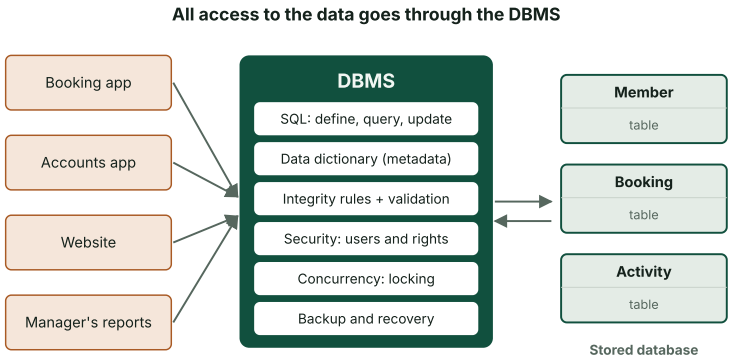
A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Data management: flat files, relational databases and the DBMS

IS01 ANSWERS

I can: Explain why flat files lead to data redundancy and data inconsistency · Describe at least five functions of a DBMS · Explain what data independence means and why it matters to an organisation



Data independence: storage can change without rewriting the programs, because they use the schema

Use this diagram to help you: Application programs reach the data only through the DBMS, which provides SQL, the data dictionary, integrity, security, concurrency control and backup; this gives data independence

A Key terms

Write the letter of the matching definition · 6 marks

A Flat file

D Data redundancy

B Data inconsistency

C Database management system (DBMS)

F Data independence

E Relational database

A A database held as a single table or file with no relationships to other tables.

B A situation where copies of the same data item hold different values, so it is unclear which is correct.

C Software that allows a database to be defined, created, queried, updated and controlled, including security, integrity and concurrent access.

D The unnecessary storage of the same data item more than once.

E A database in which data is stored in tables (relations) linked to each other through primary and foreign keys.

F The ability to change the way data is stored or structured without having to change the application programs that use it.

B Fill the gaps

Use the word bank · 5 marks

tables | DBMS | inconsistency | independence | redundancy

In a flat file the same data is often stored many times, which is called data **redundancy**. If one copy is changed and another is not, the result is data **inconsistency**. A relational database splits data into linked **tables**. The software that controls access to the database is the **DBMS**. Changing how data is stored without rewriting programs is known as data **independence**.

C True or false?

Tick one box · 6 marks

	True	False
A flat file stores data in a single table with no links to other tables.	☑	☐
Data redundancy and data inconsistency mean the same thing. *(Redundancy is duplication; inconsistency is when duplicated copies disagree. Redundancy causes the risk of inconsistency.)*	☐	☑
A DBMS keeps a data dictionary describing the structure of the database.	☑	☐
Data independence means programs must be rewritten when a field is added. *(That is program-data dependence, the problem data independence solves.)*	☐	☑
A relational database removes all redundancy completely. *(Foreign keys repeat key values on purpose; normalisation minimises redundancy rather than removing it entirely.)*	☐	☑
Record locking is one way a DBMS manages concurrent access.	☑	☐

D Exam-style questions

16 marks

1 Define the term data redundancy. [1]

> Data redundancy is the unnecessary storing of the same data more than once (e.g. in several records or files) (1). Accept: duplication of data.

2 Explain how data redundancy can lead to data inconsistency. [2]

> When the same data is held in several places, an update may be made to one copy but not the others (1) / so the copies no longer agree and the organisation cannot tell which value is correct, e.g. a member's address changed on one booking but not the rest (1).

3 Describe four functions of a database management system. [4]

> Any four from, with a description: stores, retrieves and updates data, usually via a query language such as SQL (1) / maintains the data dictionary of metadata about tables, fields, types and relationships (1) / enforces integrity: validation rules, entity and referential integrity (1) / provides security through user accounts, passwords and access rights to tables, views or fields (1) / controls concurrent access, e.g. record locking so multi-user updates do not conflict (1) / provides backup and recovery after failure (1) / provides data independence by separating programs from stored data via the schema (1).

4 Explain what is meant by data independence and give one benefit to an organisation. [3]

> Data independence means the way data is physically stored (or its logical structure) can be changed (1) / without having to change the application programs that use it, because the DBMS places a schema between programs and the data (1) / Benefit, any one: lower maintenance costs because programs need not be rewritten / faster development of new applications / e.g. a new field or index can be added and existing programs still work (1). Accept physical or logical data independence with example for the second mark.

5 Erne Leisure Centre stores all bookings in a single flat file. Discuss the benefits and drawbacks of replacing the flat file with a relational database managed by a DBMS. [6]

> Indicative content. Benefits: member details (name, address, phone) stored once in a Member table linked to Booking by MemberID, so less redundancy / an address change made in one row, so no inconsistency / DBMS enforces validation, entity and referential integrity (no booking for a non-existent member) / SQL queries can join tables to answer questions like which members booked squash and owe fees / security: access rights so reception cannot see payment details / multi-user access with record locking; backup and recovery / data independence: fields can be added without rewriting programs. Drawbacks: needs design time (ER model, normalisation) and expertise / cost of DBMS software, training and possibly a database administrator / more complex; queries need joins / migration of 4000 existing rows may introduce errors / for a very small single-user list a flat file may be adequate. Conclusion: for a leisure centre with 600 members and shared booking data, the relational approach is justified. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4):

> several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 Which problem occurs when two copies of a customer's address hold different values? [1]

A Data redundancy
B Data inconsistency ✓
C Data independence
D Referential integrity

2 Which is NOT normally a function of a DBMS? [1]

A Enforcing access rights
B Maintaining a data dictionary
C Designing the ER diagram for you ✓
D Backup and recovery

3 Adding an index without changing any application program is an example of... [1]

A logical data independence
B program-data dependence
C physical data independence ✓
D data redundancy

4 In a relational database, tables are linked using... [1]

A foreign keys ✓
B validation rules
C macros
D indexes only

5 When is a flat file most appropriate? [1]

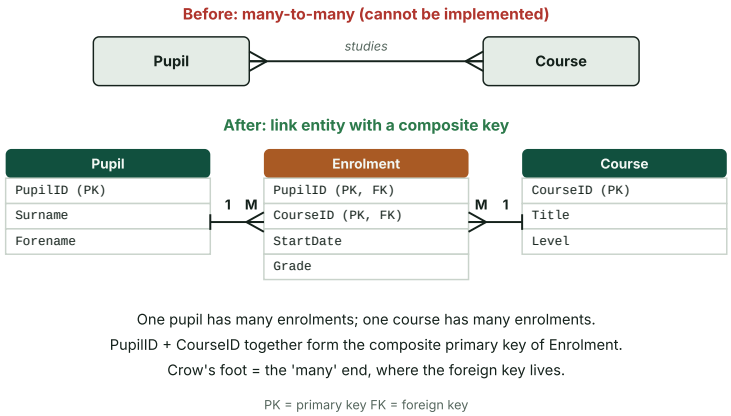
A A national bank's accounts
B A hospital patient system
C A small single-user contact list ✓
D An airline booking system

A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Entity-relationship modelling

IS02 ANSWERS

I can: Identify entities, attributes and keys from a scenario · Draw an ER diagram using crow's foot notation with the correct degree of each relationship · Resolve a many-to-many relationship using a link entity with a composite key



Use this diagram to help you: A many-to-many relationship between Pupil and Course resolved by an Enrolment link entity with a composite key, giving two one-to-many relationships

A Key terms

Write the letter of the matching definition · 6 marks

C **Entity**

B **Attribute**

A **Primary key**

D **Foreign key**

F **Composite key**

E **Link entity**

A An attribute, or combination of attributes, that uniquely identifies each record in a table.

B A property or characteristic of an entity; it becomes a field in a table.

C An object, person, place or event about which data is stored; it becomes a table in a relational database.

D An attribute in one table that is the primary key of another table and is used to link the two tables.

E An entity introduced to resolve a many-to-many relationship into two one-to-many relationships.

F A primary key made up of two or more attributes which together uniquely identify a record.

B Fill the gaps

Use the word bank · 5 marks

link · attributes · many-to-many · composite · entity

A thing about which data is stored is called an **entity**. Its properties are called **attributes**. A key made from two or more attributes is a **composite** key. A **many-to-many** relationship must be resolved by adding a **link** entity.

C True or false?

Tick one box · 5 marks

	True	False
In a one-to-many relationship the foreign key goes in the table at the 'many' end.	☑	☐

Entity names in an ER diagram should be plural, e.g. Customers. *(Entities are named as singular nouns, e.g. Customer.)*	☐	☑
A link entity changes one M:N relationship into two 1:M relationships.	☑	☐
A foreign key must be unique within its own table. *(A foreign key value is usually repeated – many records at the many end refer to the same parent.)*	☐	☑
A candidate key is any attribute that could uniquely identify a record.	☑	☐

D Exam-style questions

14 marks

1 State what is meant by a composite key. [1]

> A composite key is a primary key made up of two or more attributes which together uniquely identify each record, e.g. (PupilID, CourseID) in Enrolment (1).

2 Explain the purpose of a foreign key. [2]

> A foreign key is an attribute in one table that is the primary key of another table (1) / it is used to create the link (relationship) between the two tables, so related records can be joined, and allows referential integrity to be enforced (1).

3 A library allows each member to borrow many books, and over time each book is borrowed by many members. Draw an ER diagram showing how this relationship would be implemented in a relational database. [4]

> Three entities drawn: Member, Book and a link entity, e.g. Loan (1) / Member to Loan is one-to-many with the crow's foot at Loan (1) / Book to Loan is one-to-many with the crow's foot at Loan (1) / no direct many-to-many line remains between Member and Book, i.e. M:N correctly resolved (1). Diagram: Member 1 ——< Loan >—— 1 Book. Accept any sensible name for the link entity (Borrowing, Issue).

4 Write an entity description for the link entity in your answer above, identifying its key(s). [3]

> Loan(LoanID, MemberID*, BookID*, DateOut, DateDue, DateReturned) – underline LoanID. Marks: a sensible primary key identified, e.g. LoanID, or a composite key such as (MemberID, BookID, DateOut) (1) / MemberID identified as a foreign key to Member (1) / BookID identified as a foreign key to Book (1). Note: (MemberID, BookID) alone is not acceptable as the key because the same member can borrow the same book more than once; accept it if the candidate also includes a date.

5 Explain why a many-to-many relationship cannot be implemented directly in a relational database. [2]

> A field in a relational table must be atomic (hold a single value), so a table cannot hold a repeating list of linked keys in one field (1) / so a link (associative) entity is needed, turning the M:N into two one-to-many relationships with the link entity holding both foreign keys (1).

6 Identify the degree of the relationship between a Translink bus route and its stops, and justify your answer. [2]

> Many-to-many (M:N) (1) / because one route has many stops, and one stop can be served by many different routes (1).

E Quick quiz

Circle one answer · 5 marks

1 Which relationship needs a link entity? [1]

A One-to-one
B One-to-many
C Many-to-many ✓
D None of these

2 Enrolment(PupilID, CourseID, Grade) uses PupilID and CourseID together as... [1]

A a secondary key
B a composite key ✓
C a candidate attribute
D an index only

3 Where is the crow's foot drawn in Customer ——< Order? [1]

A At Customer
B At Order ✓
C At both ends
D Nowhere

4 A head of department manages exactly one department, and each department has one head. This is... [1]

A 1:M
B M:N
C M:1
D 1:1 ✓

5 Which statement about a foreign key is correct? [1]

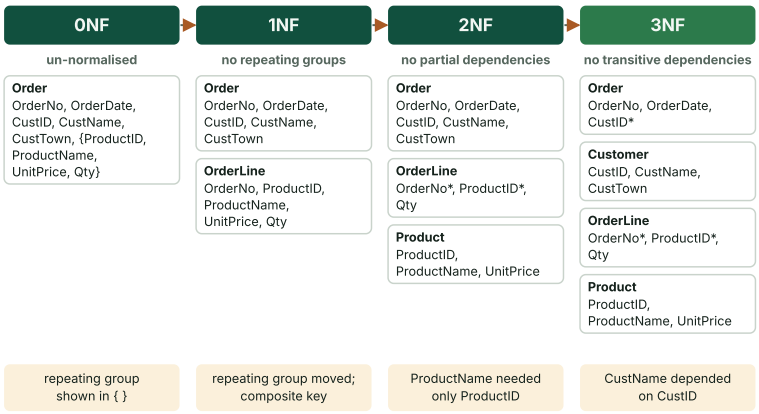
A It must be the first field in a table
B It is the primary key of another table ✓
C It cannot contain repeated values
D It is always a composite key

A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Normalisation to third normal form

IS03 ANSWERS

I can: Explain why normalisation is used and the problems (anomalies) it prevents · Take a set of data from 0NF to 3NF, showing each stage · Identify repeating groups, partial dependencies and transitive dependencies



Use this diagram to help you: Fermanagh Fresh order data normalised from 0NF through 1NF and 2NF to 3NF, showing which dependency is removed at each stage

A Key terms

Write the letter of the matching definition · 6 marks

E Normalisation

B First normal form (1NF)

D Second normal form (2NF)

A Third normal form (3NF)

F Partial dependency

C Transitive dependency

A A relation in 2NF with no transitive dependencies, so every non-key attribute depends only on the primary key.

B A relation with no repeating groups in which every attribute holds a single atomic value.

C Where a non-key attribute depends on another non-key attribute rather than directly on the primary key.

D A relation in 1NF with no partial dependencies on a composite key.

E A formal process of organising data into relations to minimise redundancy and prevent update, insertion and deletion anomalies.

F Where a non-key attribute depends on only part of a composite primary key.

B Fill the gaps

Use the word bank · 5 marks

composite | redundancy | transitive | partial | repeating

A relation is in first normal form when it has no **repeating** groups. To reach second normal form we remove **partial** dependencies, which can only exist when the table has a **composite** key. Third normal form removes **transitive** dependencies. Normalisation reduces data **redundancy**.

C True or false?

Tick one box · 6 marks

	True	False

A table with a single-attribute primary key that is in 1NF is automatically in 2NF.	☑	☐
A transitive dependency is where a non-key attribute depends on part of the key. *(That is a partial dependency. A transitive dependency is on another non-key attribute.)*	☐	☑
At 1NF the repeating group moves to a new table along with the original key.	☑	☐
Normalisation increases data redundancy. *(It reduces redundancy.)*	☐	☑
Derived values such as a line total are usually removed during normalisation because they can be calculated.	☑	☐
A fully normalised database never needs joins. *(Normalised data is spread across tables, so joins are needed to recombine it.)*	☐	☑

D Exam-style questions

18 marks

1 State two reasons why data is normalised. [2]

> Any two from: to minimise / reduce data redundancy (1) / to prevent update anomalies and so data inconsistency (1) / to prevent insertion anomalies (1) / to prevent deletion anomalies (1) / to make the database easier to maintain and change (1).

2 Define a partial dependency. [2]

> A partial dependency exists where a non-key attribute depends on only part of a composite primary key (1) / rather than on the whole key, e.g. in OrderLine(OrderNo, ProductID, ProductName, Qty) ProductName depends only on ProductID (1).

3 Define a transitive dependency and give an example. [2]

> A transitive dependency is where a non-key attribute depends on another non-key attribute rather than directly on the primary key (1) / e.g. in Order(OrderNo, OrderDate, CustID, CustName), CustName depends on CustID, which is not the key (1). Accept any correct example, e.g. TutorName depends on TutorID.

4 Student(StudentID, Name, {CourseCode, CourseTitle, TutorID, TutorName, Grade}). Normalise this relation to third normal form, showing each stage. [6]

> 1NF: remove the repeating group with a copy of the key: Student(StudentID, Name) (1) / StudentCourse(StudentID, CourseCode, CourseTitle, TutorID, TutorName, Grade) with composite key (StudentID, CourseCode) (1). 2NF: CourseTitle, TutorID and TutorName depend only on CourseCode – a partial dependency (1) / so: Enrolment(StudentID*, CourseCode*, Grade) and Course(CourseCode, CourseTitle, TutorID, TutorName) (1). 3NF: TutorName depends on TutorID, a non-key attribute – a transitive dependency (1) / so: Course(CourseCode, CourseTitle, TutorID*) and Tutor(TutorID, TutorName) (1). Final 3NF: Student(StudentID, Name), Enrolment(StudentID*, CourseCode*, Grade), Course(CourseCode, CourseTitle, TutorID*), Tutor(TutorID, TutorName). Accept the assumption that each course has one tutor; if candidate states TutorID depends on (StudentID, CourseCode) and keeps it in Enrolment with justification, credit consistently.

5 Explain what is meant by an insertion anomaly, using an example. [2]

> An insertion anomaly is when data cannot be added to the database because other, unrelated data is not yet available (1) / e.g. a new product cannot be recorded until a customer orders it, because the product details are only stored in order lines that need an OrderNo as part of the key (1).

6 Evaluate whether a database should always be normalised to third normal form. [4]

> Any four from, for a balanced evaluation: For: 3NF minimises redundancy so updates are made in one place (1) / avoids update, insertion and deletion anomalies, keeping data consistent (1) / smaller tables are easier to maintain and extend (1). Against: more tables means more joins, which can slow queries on very large databases (1) / data warehouses and reporting systems often deliberately denormalise for faster analysis (1) / some data should be kept even if it appears redundant, e.g. PriceCharged in OrderLine to record the historic price (1) / derived data may be stored for performance (1). Conclusion: operational (transaction) databases should normally be in 3NF, but denormalisation can be justified for performance or historical accuracy (1).

E Quick quiz

Circle one answer · 5 marks

1 Which form removes repeating groups? [1]

A 1NF ✓ B 2NF
C 3NF D 0NF

2 OrderLine(OrderNo, ProductID, ProductName, Qty): ProductName depends on ProductID only. This is a... [1]

A transitive dependency **B partial dependency ✓**
C repeating group D full dependency

3 Order(OrderNo, CustID, CustTown): CustTown depends on CustID. Removing this gives... [1]

A 1NF B 2NF
C 3NF ✓ D 0NF

4 Deleting the only order for a customer also loses the customer's address. This is a... [1]

A update anomaly B insertion anomaly
C referential anomaly **D deletion anomaly ✓**

5 The phrase 'the key, the whole key and nothing but the key' summarises... [1]

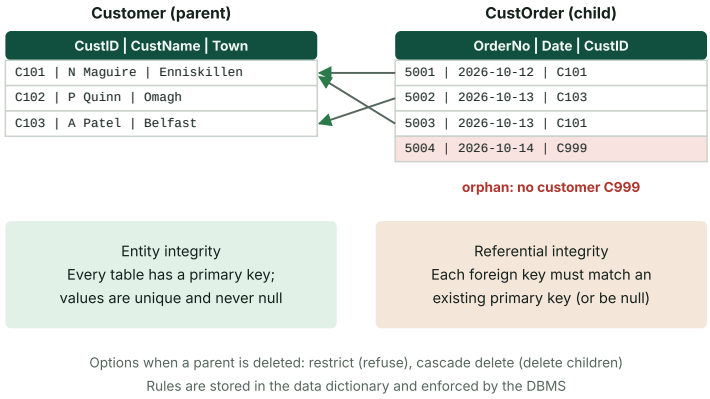
A SQL joins **B 3NF ✓**
C record locking D ER diagrams

A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Data integrity: data dictionary, integrity rules, validation and indexing

IS04 ANSWERS

I can: Describe what a data dictionary stores and who uses it · Explain entity and referential integrity, including cascade update and delete · Evaluate the use of indexes in a relational database



Use this diagram to help you: Entity integrity requires unique, non-null primary keys; referential integrity requires every foreign key to match a primary key, so an order for missing customer C999 is an orphan record

A Key terms

Write the letter of the matching definition · 6 marks

E Data dictionary

B Metadata

A Entity integrity

C Referential integrity

F Index

D Cascade delete

A The rule that every table has a primary key whose values are unique and never null.

B Data that describes other data, such as a field's name, type and size.

C The rule that every foreign key value must match an existing primary key value in the related table, preventing orphan records.

D A referential integrity option in which deleting a parent record automatically deletes its related child records.

E A central store of metadata describing every table, field, data type, key, validation rule, index and relationship in a database.

F A sorted data structure of key values and pointers that lets the DBMS locate records quickly without scanning the whole table.

B Fill the gaps

Use the word bank · 5 marks

dictionary | null | index | orphan | metadata

Data about data is called **metadata** and is stored in the data **dictionary**. A primary key must be unique and never **null**; this is entity integrity. Referential integrity prevents **orphan** records. An **index** speeds up searching but slows down updates.

C True or false?

Tick one box · 5 marks

	True	False
Referential integrity requires every foreign key value to match an existing primary key value.	☑	☐
A primary key may be left blank if the record is incomplete. *(Entity integrity forbids null primary keys.)*	☐	☑
Indexes make INSERT and UPDATE operations faster. *(They speed up searches but slow inserts and updates because the index must be updated too.)*	☐	☑
Cascade delete removes child records when their parent is deleted.	☑	☐
Verification checks that data has been copied accurately from its source.	☑	☐

D Exam-style questions

16 marks

1 State three items of metadata held in a data dictionary. [3]

> Any three from: table names / field names (1) / data types (1) / field sizes / lengths (1) / primary keys (1) / foreign keys and relationships (1) / validation rules (1) / default values (1) / indexes (1) / users and access rights (1) / descriptions / formats (1).

2 Explain what is meant by entity integrity. [2]

> Every table must have a primary key, and each primary key value must be unique (1) / and the primary key must never be null (empty), so every record can be identified unambiguously (1).

3 Explain how referential integrity would prevent an error in a database containing Customer and Order tables. [3]

> Referential integrity means every foreign key value must match an existing primary key value in the related table (1) / so the DBMS would reject an Order whose CustID does not exist in the Customer table, e.g. an order for C999 (1) / preventing orphan records; it would also stop (restrict) the deletion of a customer who still has orders, or cascade the change (1).

4 Describe the difference between cascade delete and restrict delete. [2]

> Cascade delete: when a parent record is deleted, all related child records are automatically deleted too, e.g. deleting a customer deletes their orders (1) / Restrict delete: the DBMS refuses to delete a parent record while related child records still exist (1).

5 Evaluate the use of indexes in a large relational database. [4]

> Any four from, balancing benefits and drawbacks: an index is a separate sorted structure (e.g. B-tree) of key values and pointers, so the DBMS can find records without scanning the whole table (1) / so searches, sorting and joins on indexed fields are much faster in a large database (1) / primary keys are indexed automatically; secondary indexes help on often-searched fields such as Surname (1) / but each index uses extra storage (1) / and slows INSERT, UPDATE and DELETE because every index must also be updated (1) / so index only fields used often in WHERE and JOIN, not small tables or rarely queried fields (1).

6 Explain why validation alone cannot guarantee that data is correct. [2]

> Validation only checks that data is reasonable / obeys rules, e.g. in range, correct type or format (1) / it cannot check data is actually correct – a valid but wrong value, e.g. a date of birth of 12/03/2008 instead of 21/03/2008, is accepted; verification (e.g. double entry or proofreading) is needed as well (1).

E Quick quiz

Circle one answer · 5 marks

1 Which rule stops two customers having the same CustID? [1]

A Referential integrity
B Entity integrity ✓
C Cascade update
D A format check

2 An Order with CustID C999 exists but there is no customer C999. This breaks... [1]

A entity integrity
B a range check
C referential integrity ✓
D normalisation

3 Which is NOT normally stored in a data dictionary? [1]

A Field data types
B Validation rules
C The customers' addresses ✓
D Relationships between tables

4 The main cost of adding many indexes is... [1]

A slower searches
B slower inserts and updates and more storage ✓
C loss of referential integrity
D more redundancy in tables

5 'Qty must be between 1 and 50' is a... [1]

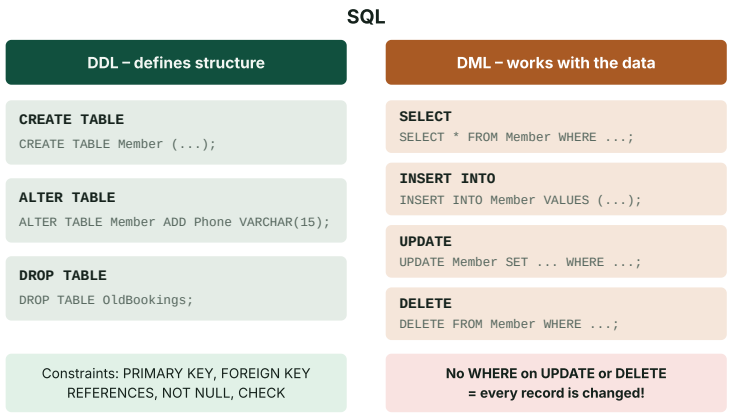
A lookup check
B presence check
C format check
D range check ✓

A2 1: INFORMATION SYSTEMS · SQL AND DATABASE MANAGEMENT

SQL data definition and data manipulation

IS05 ANSWERS

I can: Write CREATE TABLE statements with suitable data types and constraints · Write INSERT, UPDATE and DELETE statements correctly · Explain the danger of UPDATE and DELETE without a WHERE clause



Use this diagram to help you: SQL split into data definition language for structure and data manipulation language for data, with an example of each command

A Key terms

Write the letter of the matching definition · 4 marks

C SQL

B Data definition language (DDL)

D Data manipulation language (DML)

A Constraint

A A rule defined on a table, such as PRIMARY KEY, FOREIGN KEY, NOT NULL or CHECK, that the DBMS enforces on every change.

B The part of SQL used to create and change database structure, e.g. CREATE TABLE, ALTER TABLE, DROP TABLE.

C Structured Query Language: the standard declarative language for defining, querying and manipulating relational databases.

D The part of SQL used to retrieve and change data, e.g. SELECT, INSERT, UPDATE, DELETE.

B Fill the gaps

Use the word bank · 5 marks

quotes | manipulation | REFERENCES | WHERE | definition

CREATE TABLE is part of the data **definition** language. INSERT, UPDATE and DELETE are data **manipulation** commands. The **WHERE** clause chooses which records an UPDATE affects. The FOREIGN KEY constraint uses the keyword **REFERENCES** to name the parent table. Text values in SQL are written inside single **quotes** .

C True or false?

Tick one box · 5 marks

	True	False
DELETE FROM Customer; with no WHERE clause deletes every customer record.	☑	☐

SELECT is a data definition language command. *(SELECT is DML; it retrieves data rather than defining structure.)*	☐	☑
VARCHAR(30) stores text of up to 30 characters.	☑	☐
A child table should be created before the parent table it refers to. *(The parent must exist first so the FOREIGN KEY can reference it.)*	☐	☑
UPDATE changes values in existing records; it does not add new records.	☑	☐

D Exam-style questions

16 marks

1 State the difference between DDL and DML, giving one SQL command for each. [2]

> DDL (data definition language) defines / changes the structure of the database, e.g. CREATE TABLE, ALTER TABLE or DROP TABLE (1) / DML (data manipulation language) works with the data itself, e.g. SELECT, INSERT, UPDATE or DELETE (1).

2 Write an SQL statement to create a table Member with fields MemberID (6 characters, primary key), Surname (up to 30 characters, required) and JoinDate (date). [4]

> Model answer:
> ```
> CREATE TABLE Member (
>  MemberID CHAR(6) NOT NULL,
>  Surname  VARCHAR(30) NOT NULL,
>  JoinDate DATE,
>  PRIMARY KEY (MemberID)
> );
> ```
> Marks: CREATE TABLE Member with fields in brackets, correct syntax (1) / MemberID CHAR(6) (accept VARCHAR(6)) (1) / Surname VARCHAR(30) NOT NULL (1) / JoinDate DATE and PRIMARY KEY (MemberID) (1).

3 Write an SQL statement to add a member with ID M0042, surname Corrigan, who joined on 1 September 2026. [2]

> Model answer:
> ```
> INSERT INTO Member (MemberID, Surname, JoinDate)
> VALUES ('M0042', 'Corrigan', '2026-09-01');
> ```
> Marks: INSERT INTO Member with field list (accept omitted field list if values are in table order) (1) / VALUES with text and date in quotes in matching order (1). Accept #01/09/2026# (Access) or '01/09/2026'.

4 Write an SQL statement to increase the price of every product in the category 'Fruit' by 10%. [3]

> Model answer:
> ```
> UPDATE Product
> SET UnitPrice = UnitPrice * 1.1
> WHERE Category = 'Fruit';
> ```
> Marks: UPDATE Product (1) / SET UnitPrice = UnitPrice * 1.1 (accept + UnitPrice * 0.1) (1) / WHERE Category = 'Fruit' (1). Accept Price for UnitPrice.

5 A user runs DELETE FROM Booking; Explain the effect and how the database could have been protected. [3]

> Effect: every record in the Booking table is deleted because there is no WHERE clause (the table structure remains) (1) / Protection, any two from: restrict access rights so only authorised users (e.g. DBA) can run DELETE on Booking (1) / run it inside a transaction so it can be rolled back before COMMIT (1) / regular backups so the data can be restored (1) / referential integrity set to restrict deletion where related records exist (1) / audit logs to identify the user (1).

6 Explain how the FOREIGN KEY constraint in a CREATE TABLE statement helps maintain data integrity. [2]

> The FOREIGN KEY … REFERENCES clause enforces referential integrity: a value can only be entered in the foreign key field if it matches an existing primary key in the referenced (parent) table (1) / so orphan records cannot be created, e.g. no order for a customer who does not exist, and a parent with related children cannot be deleted (unless cascade is set) (1).

E Quick quiz

Circle one answer · 5 marks

1 Which SQL command adds a new record? [1]

A UPDATE
B ALTER
C INSERT INTO ✓
D CREATE

2 Which data type is most suitable for a price such as 12.49? [1]

A INT
B DECIMAL(6,2) ✓
C CHAR(5)
D BOOLEAN

3 What does ALTER TABLE do? [1]

A Changes the structure of an existing table ✓
B Changes values in records
C Deletes a table
D Sorts a table

4 Which clause of UPDATE says what the new values are? [1]

A WHERE
B VALUES
C FROM
D SET ✓

5 Which constraint acts as a presence check? [1]

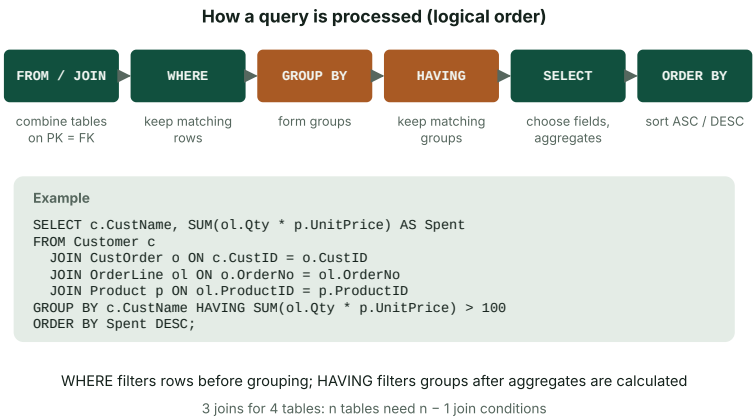
A CHECK
B NOT NULL ✓
C REFERENCES
D UNIQUE

A2 1: INFORMATION SYSTEMS · SQL AND DATABASE MANAGEMENT

SQL queries: SELECT, aggregates, grouping and joins

IS06 ANSWERS

I can: Write single-table SELECT queries with conditions and sorting · Use COUNT, SUM, AVG, MIN and MAX with GROUP BY and HAVING · Join two or three tables using primary key–foreign key matches



Use this diagram to help you: The order in which a DBMS logically processes an SQL query: FROM and JOIN, WHERE, GROUP BY, HAVING, SELECT, then ORDER BY

A Key terms

Write the letter of the matching definition · 5 marks

A Aggregate function

D GROUP BY

B HAVING

E Join

C Wildcard

A An SQL function that calculates a single value from a set of rows, such as COUNT, SUM, AVG, MIN or MAX.

B An SQL clause that filters groups after aggregation, in the way WHERE filters individual rows.

C A symbol used with LIKE to match unknown characters: % for any sequence and _ for exactly one character.

D An SQL clause that groups rows with the same value in a field so aggregate functions are calculated for each group.

E An operation that combines rows from two or more tables by matching a foreign key with the related primary key.

B Fill the gaps

Use the word bank · 5 marks

% | aggregate | HAVING | join | WHERE

The WHERE clause filters rows before grouping, while HAVING filters groups. In a LIKE pattern, the % wildcard matches any number of characters. COUNT and AVG are examples of aggregate functions. Matching a foreign key to its primary key across tables is called a join.

C True or false?

Tick one box · 5 marks

	True	False

BETWEEN 10 AND 20 includes the values 10 and 20.	☑	☐
ORDER BY sorts in descending order unless ASC is given. *(The default is ascending; DESC must be stated.)*	☐	☑
Joining three tables requires at least two join conditions.	☑	☐
WHERE COUNT(*) > 3 is the correct way to filter groups. *(Aggregates cannot be used in WHERE; use HAVING.)*	☐	☑
In LIKE patterns, _ matches exactly one character.	☑	☐

D Exam-style questions

18 marks

1 Write an SQL query to list the Surname and Forename of all pupils in YearGroup 14, in alphabetical order of Surname. [3]

> Model answer:
> SELECT Surname, Forename
> FROM Pupil
> WHERE YearGroup = 14
> ORDER BY Surname;
> Marks: SELECT Surname, Forename FROM Pupil (1) / WHERE YearGroup = 14 (1) / ORDER BY Surname (ASC) (1). Accept any sensible table name.

2 Write an SQL query to display all products whose name begins with 'Organic' and whose price is between £1 and £3. [3]

> Model answer:
> SELECT *
> FROM Product
> WHERE ProductName LIKE 'Organic%'
> AND UnitPrice BETWEEN 1 AND 3;
> Marks: SELECT * FROM Product (1) / ProductName LIKE 'Organic%' (accept * wildcard in Access) (1) / AND UnitPrice BETWEEN 1 AND 3 (accept >= 1 AND <= 3) (1).

3 Explain the difference between WHERE and HAVING. [2]

> WHERE filters individual rows before they are grouped / aggregated (1) / HAVING filters groups after GROUP BY and aggregation, so it can use aggregate functions such as COUNT(*) (1).

4 Write an SQL query to show each customer town and the number of customers in it, only for towns with more than five customers. [4]

> Model answer:
> SELECT CustTown, COUNT(*) AS NumCustomers
> FROM Customer
> GROUP BY CustTown
> HAVING COUNT(*) > 5;
> Marks: SELECT CustTown (1) / COUNT(*) (accept COUNT(CustID)) (1) / GROUP BY CustTown (1) / HAVING COUNT(*) > 5 (1).

5 Using tables Customer(CustID, CustName), CustOrder(OrderNo, OrderDate, CustID) and OrderLine(OrderNo, ProductID, Qty), write a query to list the names of customers who ordered product 'P07', with the order date. [5]

> Model answer:
> SELECT Customer.CustName, CustOrder.OrderDate
> FROM Customer
> INNER JOIN CustOrder ON Customer.CustID = CustOrder.CustID
> INNER JOIN OrderLine ON CustOrder.OrderNo = OrderLine.OrderNo

WHERE OrderLine.ProductID = 'P07';
Marks: SELECT CustName, OrderDate (1) / FROM all three tables (1) / join condition Customer.CustID = CustOrder.CustID (1) / join condition CustOrder.OrderNo = OrderLine.OrderNo (1) / WHERE ProductID = 'P07' (1). Accept the WHERE-join style (FROM Customer, CustOrder, OrderLine WHERE ... AND ...) and table aliases.

6 State what is returned if the join condition is omitted when querying two tables. [1]

A Cartesian product (cross join): every row of the first table is combined with every row of the second table (1).

E Quick quiz

Circle one answer · 5 marks

1 Which query finds surnames starting with Mc? [1]

A WHERE Surname = 'Mc%'
B WHERE Surname LIKE 'Mc%' ✓
C WHERE Surname LIKE '%Mc'
D WHERE Surname IN 'Mc'

2 Which function returns the number of rows? [1]

A SUM
B AVG
C MAX
D COUNT ✓

3 Which clause produces one result row per house? [1]

A ORDER BY House
B GROUP BY House ✓
C HAVING House
D WHERE House

4 Customer and CustOrder are joined on... [1]

A Customer.CustID = CustOrder.CustID ✓
B Customer.CustName = CustOrder.OrderNo
C any field with the same type
D no condition is needed

5 Which clause order is correct? [1]

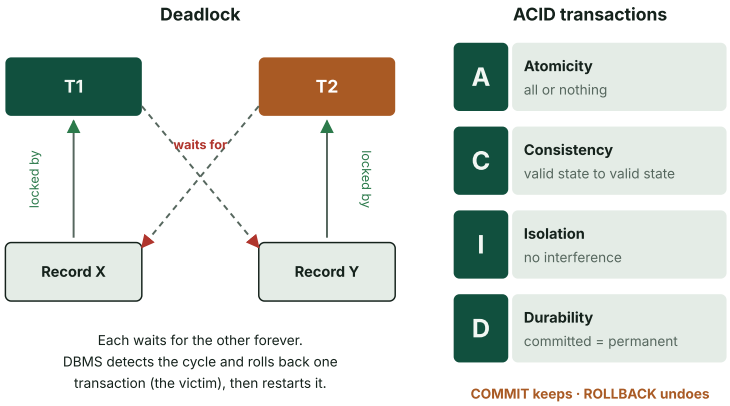
A SELECT, WHERE, FROM, ORDER BY
B SELECT, FROM, GROUP BY, WHERE
C SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY ✓
D FROM, SELECT, HAVING, WHERE

A2 1: INFORMATION SYSTEMS · SQL AND DATABASE MANAGEMENT

Multi-user databases: transactions, ACID, locking and distributed databases

IS07 ANSWERS

I can: Explain each ACID property with an example · Describe how locking prevents the lost update problem and how deadlock can arise and be resolved · Evaluate client–server and distributed database architectures



Use this diagram to help you: Deadlock: transaction T1 holds a lock on record X and waits for Y while T2 holds Y and waits for X; the DBMS rolls one back. ACID properties listed alongside

A Key terms

Write the letter of the matching definition · 6 marks

B Transaction

E ACID

C Record locking

D Deadlock

F Rollback

A Distributed database

A A single logical database whose data is stored at several sites linked by a network but appears to users as one database.

B A sequence of database operations treated as a single logical unit that must either complete entirely or not at all.

C A concurrency control technique that prevents other users updating a record while one transaction is using it.

D A situation where two or more transactions each hold a lock the other needs, so none can proceed.

E Atomicity, consistency, isolation and durability: the properties that guarantee database transactions are processed reliably.

F Undoing all changes made by a transaction since it began, returning the database to its previous consistent state.

B Fill the gaps

Use the word bank · 5 marks

Atomicity | committed | ROLLBACK | deadlock | transaction

A **transaction** is treated as one unit of work. **Atomicity** means all of it happens or none of it does. Once a transaction is **committed** its changes are permanent. Two transactions waiting for each other's locks are in **deadlock**. **ROLLBACK** undoes a transaction's changes.

C True or false?

Tick one box · 5 marks

	True	False
Isolation means concurrent transactions do not interfere with each other.	☑	☐
Durability means a transaction can be undone at any time after it is committed. *(Durability means committed changes are permanent and survive failures.)*	☐	☑
Record locking can cause deadlock.	☑	☐
A distributed database has a single point of failure. *(Data is spread across sites, so failure of one site does not stop the others.)*	☐	☑
In a client–server database only the results of a query are sent back to the client.	☑	☐

D Exam-style questions

21 marks

1 Define the term transaction in the context of databases. [2]

> A transaction is a group of one or more database operations (e.g. updates) (1) / that must be treated as a single logical unit of work: either all succeed (COMMIT) or none are kept (ROLLBACK) (1).

2 Explain atomicity and durability, using a bank transfer as an example. [4]

> Atomicity: all operations in the transaction happen or none do (1) / e.g. the £50 is never debited from savings without also being credited to the current account; if the credit fails the debit is rolled back (1) / Durability: once the transaction is committed its changes are permanent (1) / e.g. the transfer is still recorded after a power cut or server crash/restart, because it was written to the log/disk (1).

3 Describe how the lost update problem can occur and how record locking prevents it. [4]

> Two users / transactions read the same record at the same time, e.g. both read Balance = 100 (1) / each changes their copy and writes it back, e.g. A writes 70 and then B writes 80, so A's update is overwritten (lost); the balance should be 50 (1) / With record locking, the first transaction to read the record for update locks it (exclusive lock) (1) / the second must wait until the lock is released at COMMIT/ROLLBACK, so it reads the updated value (70) and writes 50 (1).

4 Explain what is meant by deadlock and how a DBMS can resolve it. [3]

> Deadlock is when two (or more) transactions each hold a lock on a record the other needs, e.g. T1 locks X and waits for Y while T2 locks Y and waits for X (1) / so neither can ever continue (1) / the DBMS detects the cycle and chooses one transaction as a victim, rolls it back to release its locks and restarts it later (1). Accept prevention: lock all records at the start or always lock in the same order; timestamp ordering.

5 Explain the purpose of COMMIT and ROLLBACK. [2]

> COMMIT makes all the changes made by the transaction permanent in the database (1) / ROLLBACK undoes all changes made since the transaction began, returning the database to its previous consistent state, e.g. after an error (1).

6 Discuss the advantages and disadvantages of a distributed database for a chain of cinemas across Northern Ireland. [6]

> Indicative content. Advantages: each cinema (e.g. Enniskillen, Belfast, Derry) holds its own bookings and screenings locally, so access is fast and less dependent on the WAN / no single point of failure – if one site or link fails the others keep selling tickets / replication allows head office to see data from all sites; data closer to where it is used / scalable – a new cinema can be added as a new site / local control of local data.
> Disadvantages: complex to keep replicated data consistent, e.g. film schedules or loyalty points changed at one site / distributed transactions need two-phase commit, slower and complex / more sites to secure, so security is harder / higher costs: hardware, network links and specialist staff / queries across all sites (e.g. company-wide sales) can be slower. Conclusion: likely worthwhile for a multi-site chain if strong network links and replication control are in place; a central client–server database with good links may be a simpler alternative. Level 3 (5–6):

> balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 Which ACID property ensures a debit is never made without its matching credit? [1]

A Isolation
B Durability
C Atomicity ✓
D Consistency

2 User A's update is overwritten by user B's update. This is the... [1]

A deadlock problem
B lost update problem ✓
C Cartesian product
D update anomaly

3 How does a DBMS usually resolve deadlock? [1]

A Deletes both records
B Rolls back one transaction ✓
C Commits both transactions
D Turns off locking

4 Storing copies of the same data at several sites is... [1]

A fragmentation
B normalisation
C indexing
D replication ✓

5 Which statement makes all changes in a transaction permanent? [1]

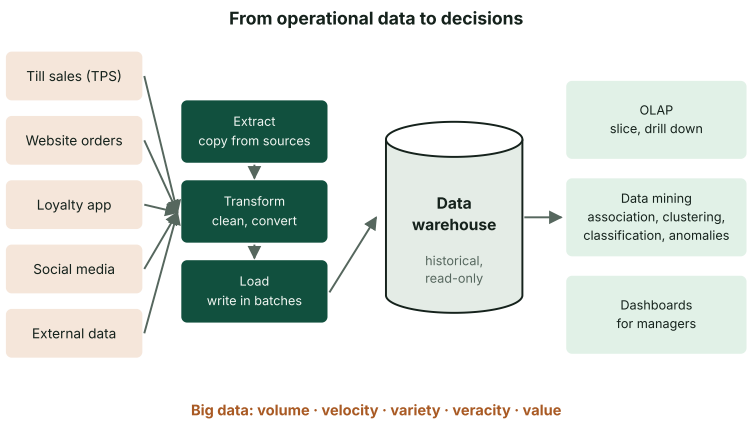
A COMMIT ✓
B ROLLBACK
C LOCK
D SELECT

A2 1: INFORMATION SYSTEMS · SQL AND DATABASE MANAGEMENT

Big data, data warehousing, data mining and analytics

IS08 ANSWERS

I can: Describe big data using volume, velocity, variety, veracity and value · Explain how a data warehouse is built and how it differs from an operational database · Evaluate the benefits of data mining for an organisation against privacy concerns



Use this diagram to help you: Data from operational sources is extracted, transformed and loaded into a data warehouse, then analysed with OLAP and data mining to support decisions

A Key terms

Write the letter of the matching definition · 5 marks

A Big data

D Data warehouse

B ETL

E Data mining

C OLAP

A Data sets so large, fast-moving or varied that they cannot be processed efficiently with traditional database tools.

B Extract, transform, load: the process of copying data from source systems, cleaning and converting it, and loading it into a data warehouse.

C Online analytical processing: tools that let users analyse multidimensional data, for example by slicing and drilling down.

D A large store of integrated, historical, read-only data from many sources, organised for analysis and decision-making.

E The process of analysing large data sets to discover previously unknown patterns, relationships and trends.

B Fill the gaps

Use the word bank · 5 marks

clustering | association | transform | warehouse | volume

Big data is often described by its **volume**, velocity and variety. A data **warehouse** stores historical data from many sources for analysis. Data is loaded using extract, **transform**, load. Discovering that customers who buy one item also buy another is called **association**. Finding natural groups of customers is called **clustering**.

C True or false?

Tick one box · 5 marks

	True	False
Velocity refers to the speed at which big data is generated and processed.	☑	☐
Data in a warehouse is normally updated many times a second by staff. *(Warehouse data is historical and non-volatile; it is loaded in batches and read for analysis.)*	☐	☑
Data mining can discover patterns that were not previously known.	☑	☐
Anonymised data can never be linked back to an individual. *(Combining data sets can re-identify people.)*	☐	☑
Data warehouses are often denormalised to speed up analytical queries.	☑	☐

D Exam-style questions

20 marks

1 Describe three characteristics of big data. [3]

> Any three from, each described: Volume – huge quantities of data, terabytes to petabytes (1) / Velocity – data arrives very quickly and must be processed fast, often in real time (1) / Variety – many types and formats, structured and unstructured such as text, images and video (1) / Veracity – data may be of uncertain accuracy or trustworthiness (1) / Value – useful insight can be gained from it (1).

2 Explain the stages of ETL used when loading a data warehouse. [3]

> Extract: data is copied from the various operational source systems (1) / Transform: data is cleaned and made consistent, e.g. converting formats, removing duplicates and errors (1) / Load: the transformed data is loaded into the data warehouse for analysis (1).

3 Give two differences between an operational database and a data warehouse. [2]

> Any two from: operational databases support daily transactions, a warehouse supports analysis and decision-making (1) / operational data is current and changing, warehouse data is historical and read-only (non-volatile) (1) / operational databases are normalised to 3NF, warehouses are often denormalised (e.g. star schema) (1) / operational queries are many short updates, warehouse queries are few, large, complex reads (1) / operational users are staff, warehouse users are managers/analysts (1) / warehouse integrates data from many sources (1).

4 Describe two data mining techniques a supermarket could use, with an example of each. [4]

> Any two techniques, each with description (1) and supermarket example (1): Association / market basket analysis – finding items bought together (1) / e.g. customers who buy nappies also buy beer, so they are placed nearby or offered together (1) / Clustering – finding natural groups not known in advance (1) / e.g. segmenting shoppers into 'young families' or 'budget shoppers' for targeted offers (1) / Classification – assigning records to known groups (1) / e.g. predicting whether a loyalty member is likely to switch to a competitor (1) / Anomaly detection – spotting unusual patterns (1) / e.g. detecting fraudulent card or voucher use (1) / Forecasting / regression (1) / e.g. predicting demand for barbecue food when hot weather is forecast (1).

5 Discuss the benefits and privacy issues of a supermarket using loyalty-card data for data mining. [8]

> Indicative content. Benefits: discovers buying patterns (association, clustering) for targeted, personalised offers, increasing sales and loyalty / better stock control and demand forecasting, reducing waste / store layout and promotions planned with evidence / customers get relevant discounts / competitive advantage. Privacy issues: data collected for loyalty points may be reused for other purposes, breaching purpose limitation under UK GDPR / profiling can reveal sensitive information (e.g. pregnancy, health, religion inferred from purchases – special category data) / customers may not understand what they consented to; transparency needed / data may be shared or sold to third parties / combining anonymised data sets can re-identify individuals / large stores of personal data are targets for hackers; a breach could lead to Information Commission (formerly the ICO) enforcement and fines / data minimisation and storage limitation principles must be followed; customers have rights to access, object to direct marketing and erasure. Conclusion: benefits are significant, but only if processing is lawful, transparent, secure and proportionate. Level 3 (6–8): thorough, balanced discussion of both sides, clearly applied to the context, with a justified conclusion; QWC: fluent, well structured, accurate

specialist terms. Level 2 (3–5): a range of relevant points with some explanation and application; may be unbalanced or lack a conclusion; QWC: generally clear. Level 1 (1–2): limited, mostly descriptive points with little application; QWC: basic, errors may obscure meaning.

E Quick quiz

Circle one answer · 5 marks

1 Which V describes the trustworthiness of data? [1]

A Volume
B Velocity
C Veracity ✓
D Variety

2 What does the T in ETL stand for? [1]

A Transmit
B Transform ✓
C Transfer
D Test

3 Finding unusual card transactions that may be fraud is... [1]

A anomaly detection ✓
B association
C normalisation
D replication

4 Which is a feature of a data warehouse? [1]

A Current data only
B Fully normalised for updates
C Used for daily order entry
D Historical, integrated data for analysis ✓

5 'What should we do?' is answered by... [1]

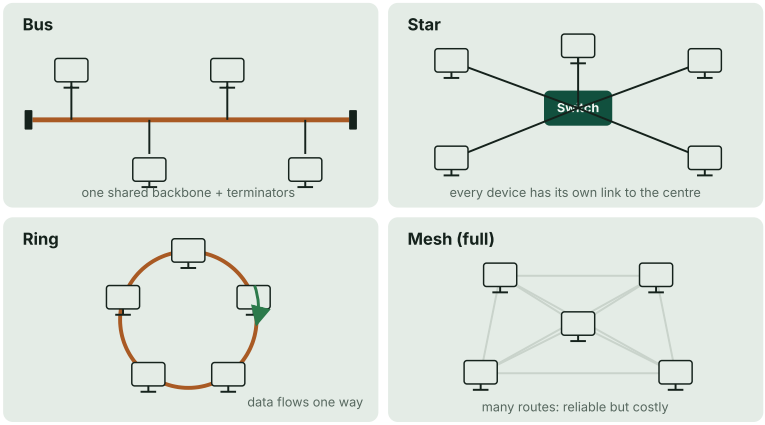
A descriptive analytics
B prescriptive analytics ✓
C an index
D a transaction log

A2 1: INFORMATION SYSTEMS · NETWORKS

Network types, topologies and hardware

IS09 ANSWERS

I can: Distinguish network types by geographical scale and ownership · Evaluate the four main topologies for a given scenario · Explain the role of each item of network hardware in moving data



Use this diagram to help you: The four network topologies: bus with a shared backbone, star with a central switch, ring as a closed loop, and mesh with multiple links

A Key terms

Write the letter of the matching definition · 6 marks

D Local area network (LAN)

C Wide area network (WAN)

E Topology

B Switch

F Router

A Gateway

A A device that connects two networks using different protocols and translates between them.

B A device that connects devices within a LAN and forwards each frame only to the port of its destination, using MAC addresses.

C A network covering a large geographical area, connecting LANs using third-party telecommunications links.

D A network covering a single site or building, with infrastructure owned and managed by the organisation.

E The physical or logical arrangement of the devices and links in a network.

F A device that connects different networks and forwards packets towards their destination using IP addresses and routing tables.

B Fill the gaps

Use the word bank · 5 marks

star | router | WAN | MAC | LAN

A network covering a single building is a **LAN**. A network spanning several countries is a **WAN**. In a **star** topology every device connects to a central switch. A **router** forwards packets between different networks using IP addresses. Every NIC has a unique **MAC** address.

C True or false?

Tick one box · 5 marks

	True	False
A switch forwards frames only to the port of the destination device.	☑	☐
In a bus topology a break in the backbone affects only one device. *(A backbone break stops the whole bus network.)*	☐	☑
A PAN typically connects a person's own devices over a few metres.	☑	☐
Routers use MAC addresses to forward packets across the internet. *(Routers use IP addresses; switches use MAC addresses.)*	☐	☑
A full mesh gives every node a direct link to every other node.	☑	☐

D Exam-style questions

14 marks

1 State one difference between a LAN and a WAN. [1]

> Any one from: a LAN covers a small area such as one site or building, while a WAN covers a large geographical area such as countries (1) / a LAN's infrastructure is owned by the organisation, while a WAN uses third-party telecoms links / leased lines (1).

2 Describe the function of a switch in a star network. [2]

> The switch connects all the devices in the star; each device has its own cable to it (1) / it learns the MAC address of each device and forwards each frame only to the port of the destination device, rather than to all devices, reducing traffic and collisions (1).

3 Explain the difference between a router and a gateway. [2]

> A router connects different networks (e.g. a LAN to the internet) and forwards packets using IP addresses and routing tables to choose the best route (1) / a gateway connects networks that use different protocols, translating between them (1).

4 Give two advantages and one disadvantage of a mesh topology. [3]

> Advantages, any two: very reliable – if one link fails data can be rerouted by another path (1) / no single point of failure (1) / high performance as traffic can travel on many routes / direct links (1) / easy to diagnose faults / secure point-to-point links (1). Disadvantage, any one: expensive because of the large amount of cabling / connections (1) / complex to install, configure and manage (1).

5 A hotel is choosing between a bus and a star topology for its new network. Evaluate both options and recommend one. [6]

> Indicative content. Bus: all devices share one backbone cable with terminators / cheap – less cable, easy to extend for a small network / but only one device can transmit at a time, so collisions increase with traffic and performance falls as guests and staff use the network / a break in the backbone stops the whole network / poor security as all data passes every device / hard to locate faults. Star: each device has its own cable to a central switch / a cable fault affects only one device (fault isolation) / switch sends data only to the destination, so high performance even with many users / easy to add devices and to segment staff and guest networks / but more cable, so more expensive / the switch is a single point of failure (can be mitigated with a spare). Recommendation: star, because a hotel has many users (rooms, reception, bookings, payments), needs reliability and performance, and the extra cabling cost is justified. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 Which network type covers a city? [1]

A PAN
B LAN
C MAN ✓
D WAN

2 Which device lets wireless devices join a wired LAN? [1]

A Wireless access point ✓ B Gateway

C Hub D Repeater

3 In which topology does data travel around a loop from node to node? [1]

A Bus B Star

C Mesh **D Ring ✓**

4 What is the main weakness of a star topology? [1]

A Collisions on the backbone **B The central switch is a single point of failure ✓**

C It cannot be extended D No device has a MAC address

5 Which device translates between networks using different protocols? [1]

A Switch B NIC

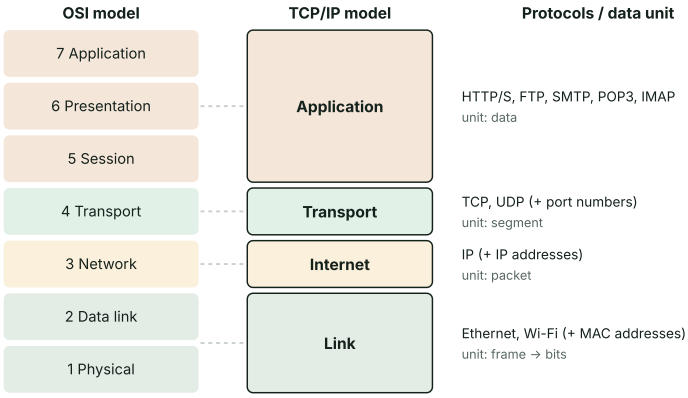
C Gateway ✓ D WAP

A2 1: INFORMATION SYSTEMS · NETWORKS

Layered network models and protocols

IS10 ANSWERS

I can: Explain the benefits of a layered approach to network design · Map the OSI model to the TCP/IP model and place protocols in the correct layer · Compare TCP and UDP and choose an appropriate one for an application



Sending: down the layers, each adds a header (encapsulation). Receiving: up the layers, headers removed.

Use this diagram to help you: The seven OSI layers mapped to the four TCP/IP layers, with example protocols and the header added at each layer during encapsulation

A Key terms

Write the letter of the matching definition · 6 marks

C **Protocol**

D **OSI model**

E **TCP/IP model**

B **Encapsulation (networking)**

F **TCP**

A **UDP**

A User Datagram Protocol: a connectionless transport protocol that is fast but does not guarantee delivery or order.

B The process by which each layer adds its own header to data as it passes down the layers before transmission.

C A set of rules that governs how data is formatted, transmitted and received so devices can communicate.

D A seven-layer reference model of network communication: physical, data link, network, transport, session, presentation and application.

E The four-layer model used by the internet: link, internet, transport and application.

F Transmission Control Protocol: a connection-oriented transport protocol giving reliable, ordered delivery using acknowledgements and retransmission.

B Fill the gaps

Use the word bank · 5 marks

encapsulation | TCP | seven | UDP | protocol

A **protocol** is a set of rules for communication. The OSI model has **seven** layers. Adding a header at each layer is called **encapsulation**. **TCP** guarantees reliable, ordered delivery, while **UDP** is faster but connectionless.

C True or false?

Tick one box · 5 marks

	True	False
HTTPS uses TLS to encrypt data between the browser and the web server.	☑	☐
IMAP downloads email to one device and deletes it from the server. *(That is POP3; IMAP keeps mail on the server and synchronises devices.)*	☐	☑
The TCP/IP model has four layers.	☑	☐
UDP retransmits lost segments. *(UDP has no acknowledgements or retransmission.)*	☐	☑
IP operates at the internet (network) layer.	☑	☐

D Exam-style questions

16 marks

1 Define the term protocol. [1]

> A protocol is a set of rules that governs how data is formatted, transmitted and received between devices (1).

2 Explain two advantages of using a layered model for network communication. [4]

> Any two advantages, each stated (1) and explained (1): one layer can be changed or replaced without affecting the others (1) / e.g. Wi-Fi can replace Ethernet at the link layer without changing TCP or HTTP (1) / complex communication is split into smaller, manageable tasks (1) / so each layer is easier to design, develop, teach and troubleshoot (1) / layers follow open standards (1) / so equipment and software from different manufacturers can interoperate (1).

3 Name the four layers of the TCP/IP model in order from the top. [2]

> Application, Transport (1) / Internet (accept Network), Link (accept Network access / Network interface) (1). All four in the correct order from the top for both marks; award 1 mark if all four are named but the order is wrong.

4 Describe the difference between POP3 and IMAP. [2]

> POP3 downloads email from the server to one device, usually deleting it from the server (1) / IMAP keeps email on the server and synchronises it across multiple devices, so the same mailbox is seen on phone and laptop (1).

5 Compare TCP and UDP and state which would be more suitable for an online multiplayer game, with a reason. [4]

> Any three comparison points: TCP is connection-oriented (handshake first), UDP is connectionless (1) / TCP guarantees delivery using acknowledgements and retransmission, UDP does not (1) / TCP reassembles segments in order using sequence numbers, UDP packets may arrive out of order (1) / UDP has a smaller header / less overhead, so it is faster with lower latency (1). Suitability: UDP for an online game, because low latency matters more than occasional lost packets; retransmitting old position data would arrive too late (1).

6 Describe what happens to data as it passes down the layers on a sending device. [3]

> Data moves down from the application layer through each layer in turn (1) / each layer adds its own header (and sometimes trailer) – encapsulation (1) / e.g. transport adds ports and sequence numbers to make segments, internet/network adds source and destination IP addresses to make packets, link adds MAC addresses to make frames, which are sent as bits/signals (1).

E Quick quiz

Circle one answer · 5 marks

1 Which protocol sends email between mail servers? [1]

A IMAP
B POP3
C SMTP ✓
D FTP

2 Which OSI layer deals with MAC addresses and frames? [1]

A Data link ✓
B Network
C Transport
D Session

3 Live voice calls usually use UDP because... [1]

A it guarantees delivery

B low delay matters more than the odd lost packet ✓

C it encrypts data

D it uses MAC addresses

4 Which TCP/IP layer adds port numbers? [1]

A Link

B Internet

C Application

D Transport ✓

5 A key benefit of layering is that... [1]

A it removes the need for protocols

B one layer can be changed without affecting the others ✓

C data no longer needs headers

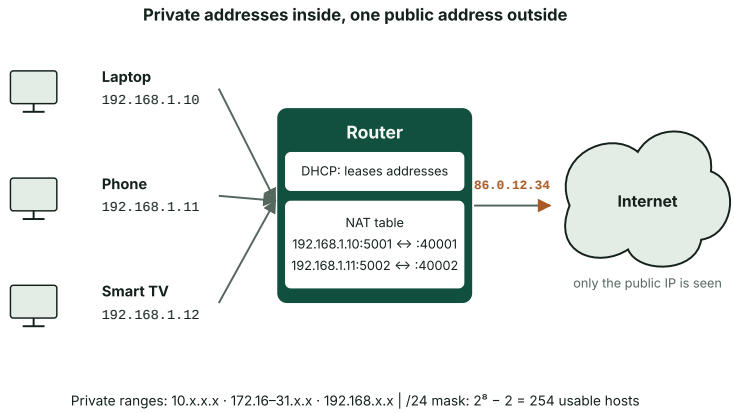
D networks need no hardware

A2 1: INFORMATION SYSTEMS · NETWORKS

IP addressing, switching and routing

IS11 ANSWERS

I can: Explain the structure of IPv4 and IPv6 addresses and why IPv6 was introduced · Explain how DHCP, NAT and subnetting are used in an organisation's network · Compare packet and circuit switching and describe how routers forward packets



Use this diagram to help you: A home network where DHCP gives devices private IP addresses and the router uses NAT to share one public IP address with the internet

A Key terms

Write the letter of the matching definition · 6 marks

C IP address

B Subnet mask

E DHCP

D Network address translation (NAT)

F Packet switching

A Circuit switching

A A method of transmission in which a dedicated path is set up and reserved between sender and receiver for the whole communication.

B A value that identifies which bits of an IP address form the network part and which form the host part.

C A logical numerical address that identifies a device on a network so data can be routed to it.

D A router technique that maps private IP addresses to a public IP address so many devices can share one public address.

E Dynamic Host Configuration Protocol: a service that automatically assigns IP addresses and network settings to devices.

F A method of transmission in which data is split into packets that travel independently across shared links and are reassembled at the destination.

B Fill the gaps

Use the word bank · 5 marks

DHCP | NAT | table | packets | 32

An IPv4 address is **32** bits long. Devices are automatically given an IP address by **DHCP**. **NAT** lets many devices with private addresses share one public address. On the internet, data is split into **packets** that may take different routes. Each router uses its routing **table** to choose the next hop.

C True or false?

Tick one box · 5 marks

	True	False
192.168.1.15 is a private IP address.	☑	☐
Circuit switching shares the same link among many users' packets. *(That is packet switching; circuit switching reserves a dedicated path.)*	☐	☑
A /24 network has 254 usable host addresses.	☑	☐
Packets always arrive in the order in which they were sent. *(Packets can take different routes and arrive out of order; they are reassembled using sequence numbers.)*	☐	☑
IPv6 addresses are usually written in hexadecimal.	☑	☐

D Exam-style questions

16 marks

1 State the number of bits in an IPv4 address and in an IPv6 address. [2]

> IPv4: 32 bits (1) / IPv6: 128 bits (1).

2 Explain why IPv6 was introduced. [2]

> IPv4 provides only about 4.3 billion (2^32) addresses, which have run out (1) / because of the huge growth in internet-connected devices such as smartphones and IoT devices; IPv6's 128-bit addresses give about 3.4 × 10^38 addresses (1).

3 A network uses the subnet mask 255.255.255.192 (/26). Calculate the number of usable host addresses. Show your working. [3]

> Host bits = 32 − 26 = 6 (1) / 2^6 = 64 addresses (1) / subtract network and broadcast addresses: 64 − 2 = 62 usable hosts (1).

4 Describe the purpose of DHCP. [2]

> DHCP automatically assigns network configuration to a device when it joins the network: IP address, subnet mask, default gateway and DNS server (1) / addresses are leased for a time, so no manual configuration is needed, duplicate addresses are avoided and addresses are reused (1).

5 Explain how network address translation (NAT) allows several devices in a home to access the internet. [3]

> Devices in the home are given private IP addresses (e.g. 192.168.x.x) that are not routable on the internet (1) / the router replaces each device's private source address with its single public IP address (and port) when packets go out (1) / it records each connection in a NAT table so that replies are translated back and forwarded to the correct device (1).

6 Compare packet switching and circuit switching. [4]

> Any four comparison points: circuit switching sets up a dedicated path for the whole communication; packet switching splits data into packets that may take different routes (1) / circuit switching reserves resources even when silent, wasteful; packet switching shares links between many users, efficient (1) / circuit switching delivers at a constant rate, in order; packets may arrive out of order and are reassembled using sequence numbers (1) / if a link fails a circuit-switched call is dropped; packets are rerouted around failures (1) / examples: traditional landline telephone network vs the internet (1).

E Quick quiz

Circle one answer · 5 marks

1 How many usable hosts does a /28 network have? [1]

A 16 **B 14 ✓**

C 28 D 30

2 Which service gives a laptop its IP address when it joins the school Wi-Fi? [1]

A NAT
B DNS
C DHCP ✓
D FTP

3 What is the purpose of the time to live field in a packet? [1]

A To encrypt the packet
B To stop packets circulating forever ✓
C To record the sender's MAC address
D To set the bit rate

4 Which is a feature of circuit switching? [1]

A Dedicated path for the whole call ✓
B Packets take different routes
C Links shared efficiently
D Packets reassembled using sequence numbers

5 Which address is public? [1]

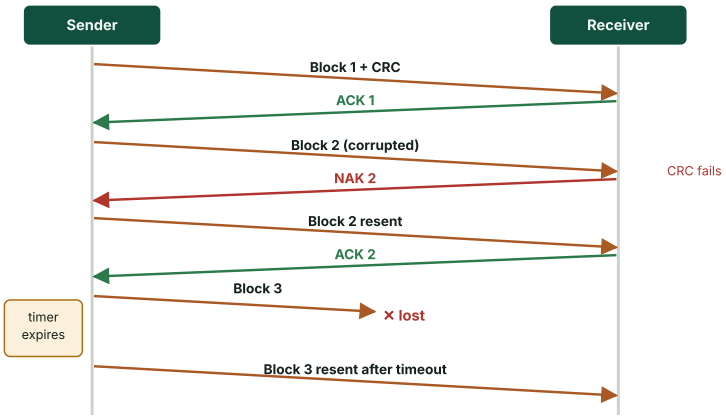
A 10.4.2.1
B 172.16.0.9
C 192.168.0.1
D 8.8.8.8 ✓

A2 1: INFORMATION SYSTEMS · NETWORKS

Data transmission and error control

IS12 ANSWERS

I can: Compare copper, fibre-optic and wireless media for a given use · Calculate transmission times using bit rate and explain the transmission modes · Apply parity and checksum methods and explain how ARQ corrects errors



Use this diagram to help you: Automatic repeat request: block 1 is acknowledged, block 2 arrives with an error and is negatively acknowledged then resent, and block 3 is lost so a timeout triggers retransmission

A Key terms

Write the letter of the matching definition · 6 marks

A Bandwidth

B Bit rate

E Full-duplex

C Asynchronous transmission

F Parity bit

D Cyclic redundancy check (CRC)

A The capacity of a communication channel (range of frequencies), which determines the maximum rate at which data can be transmitted.

B The number of bits transmitted per second over a communication channel.

C Transmission without a shared clock, where each character is framed by start and stop bits.

D An error-detection method in which a remainder calculated by dividing the data by an agreed polynomial is sent and checked by the receiver.

E A transmission mode in which data can travel in both directions at the same time.

F An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), used to detect errors.

B Fill the gaps

Use the word bank · 5 marks

ARQ · checksum · parity · full-duplex · serial

Sending bits one after another along a single wire is **serial** transmission. Communication in both directions at the same time is **full-duplex**. An extra bit added to make the number of 1s even is a **parity** bit. A **checksum** is a value calculated from a block of data and sent with it. **ARQ** corrects errors by retransmitting data.

C True or false?

Tick one box · 6 marks

	True	False
Fibre-optic cable is immune to electrical interference.	☑	☐
A walkie-talkie is an example of full-duplex communication. *(Only one person can transmit at a time, so it is half-duplex.)*	☐	☑
Parity checking can detect any single-bit error.	☑	☐
Asynchronous transmission uses a shared clock signal between sender and receiver. *(Synchronous uses a shared clock; asynchronous uses start and stop bits.)*	☐	☑
Parallel transmission suffers from skew over long distances.	☑	☐
A checksum corrects errors by itself. *(It only detects errors; correction needs retransmission (ARQ) or forward error correction.)*	☐	☑

D Exam-style questions

17 marks

1 Distinguish between bandwidth and bit rate. [2]

> Bandwidth is the range of frequencies a channel can carry, which determines its maximum data capacity (1) / bit rate is the number of bits actually transmitted per second (e.g. Mbps), which may be lower due to congestion, distance or interference (1).

2 Calculate how long it would take to transfer a 250 MB file over a 20 Mbps connection. Show your working. [3]

> Convert to bits: 250 MB × 8 = 2000 Mb (megabits) (1) / time = 2000 ÷ 20 (1) / = 100 seconds (1 minute 40 seconds) (1). Accept 104.9 s (≈105 s) if 1 MB = 1024 × 1024 bytes and 20 Mbps = 20 000 000 bps is stated.

3 Add an even parity bit to 1101001 and explain one limitation of parity checking. [3]

> 1101001 contains four 1s, which is already even (1) / so the even parity bit is 0: 11010010 (accept 01101001 if parity bit placed first) (1) / Limitation, any one: cannot detect an even number of bit errors, e.g. two bits flipped (1) / cannot identify which bit is wrong, so cannot correct the error (1).

4 Explain why serial transmission is used over long distances instead of parallel. [2]

> Parallel transmission over long distances suffers from skew – bits on different wires arrive at different times (1) / and crosstalk / interference between wires; serial sends bits one after another on one channel, so has no skew and needs fewer wires, making it cheaper and reliable over long distances (1).

5 Describe how automatic repeat request (ARQ) is used to correct transmission errors. [4]

> The receiver checks each block of data using an error-detection method such as CRC or checksum (1) / if the block is correct it sends a positive acknowledgement (ACK) to the sender (1) / if an error is found it sends a negative acknowledgement (NAK) or no acknowledgement (1) / if the sender receives a NAK, or its timer runs out (timeout) before an ACK arrives, it retransmits the block (1).

6 Recommend a transmission medium to link two buildings 400 m apart on a school campus, justifying your answer. [3]

> Fibre-optic cable (1) / copper twisted pair is limited to about 100 m per run, but fibre works over long distances such as 400 m (1) / plus any one: very high bandwidth for future needs / immune to electrical interference outdoors and between buildings (and lightning/earth differences) / hard to tap so more secure (1). Accept a point-to-point wireless (microwave/laser) link if justified, e.g. no need to dig trenches, with line of sight (max 2 marks).

E Quick quiz

Circle one answer · 5 marks

1 How long does a 100 MB file take at 50 Mbps (1 MB = 8 Mb)? [1]

A 2 s **B 16 s ✓**

C 50 s

D 8 s

2 Which medium uses pulses of light? [1]

A Coaxial

B Twisted pair

C Fibre-optic ✓

D Microwave

3 With odd parity, what is the parity bit for 1100110? [1]

A 1 ✓

B 0

C 2

D It cannot be calculated

4 In ARQ, what does the sender do if no acknowledgement arrives before the timeout? [1]

A Stops sending

B Sends the next block

C Asks the user

D Retransmits the block ✓

5 Which method detects burst errors most reliably? [1]

A Parity bit

B CRC ✓

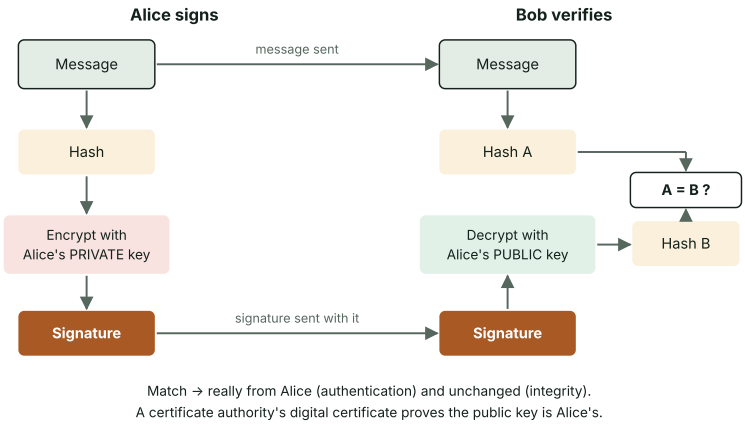
C Echo by eye

D Simplex

Network security: threats, firewalls, encryption and authentication

IS13 ANSWERS

I can: Describe common network threats and match each to suitable countermeasures · Explain how symmetric and asymmetric encryption, digital signatures and certificates work together · Evaluate firewalls, proxies, authentication methods and VPNs for an organisation



Use this diagram to help you: Creating a digital signature by hashing a message and encrypting the hash with the sender's private key, and verifying it by decrypting with the sender's public key and comparing hashes

A Key terms

Write the letter of the matching definition · 6 marks

D Firewall

B Proxy server

C Symmetric encryption

E Asymmetric encryption

F Digital signature

A Digital certificate

A An electronic document issued by a certificate authority that binds a public key to the identity of its owner.

B A server that acts as an intermediary between users and the internet, hiding internal addresses, filtering content and caching pages.

C Encryption in which the same secret key is used both to encrypt and to decrypt data.

D Hardware or software that monitors network traffic and allows or blocks it according to a set of security rules.

E Encryption using a mathematically related key pair: data encrypted with the public key can only be decrypted with the private key, and vice versa.

F A hash of a message encrypted with the sender's private key, used to verify the sender's identity and that the message has not been altered.

B Fill the gaps

Use the word bank · 5 marks

private | certificate | Symmetric | VPN | firewall

A **firewall** blocks or allows traffic according to rules. **Symmetric** encryption uses the same key to encrypt and decrypt. In asymmetric encryption, data encrypted with Bob's public key can only be

decrypted with Bob's **private** key. A digital **certificate** is issued by a certificate authority. A **VPN** creates an encrypted tunnel across the internet.

C True or false?

Tick one box · 5 marks

	True	False
A digital signature is created using the sender's private key.	☑	☐
Asymmetric encryption is faster than symmetric encryption. *(Asymmetric is much slower, so it is used mainly for key exchange and signatures.)*	☐	☑
A proxy server can cache web pages to reduce internet traffic.	☑	☐
Multi-factor authentication uses two passwords. *(MFA uses factors from different categories, e.g. a password and a phone code or fingerprint.)*	☐	☑
HTTPS uses both asymmetric and symmetric encryption.	☑	☐

D Exam-style questions

19 marks

1 Describe the purpose of a firewall. [2]

> A firewall monitors incoming and outgoing network traffic (1) / and allows or blocks it according to a set of rules, e.g. based on IP addresses, ports or protocols, to prevent unauthorised access (1).

2 Give two functions of a proxy server. [2]

> Any two from: makes requests to the internet on behalf of users, hiding internal IP addresses (1) / filters web content / blocks access to unsuitable sites (1) / caches popular web pages to speed up access and save bandwidth (1) / logs users' web activity (1).

3 Explain the key distribution problem in symmetric encryption and how asymmetric encryption overcomes it. [3]

> In symmetric encryption the same secret key is used to encrypt and decrypt, so the key must somehow be shared with the receiver (1) / if it is sent over the network it could be intercepted, and anyone with it can read the messages (1) / asymmetric encryption uses a key pair: the public key can be published openly and used to encrypt, but only the matching private key, which is never shared, can decrypt (1). Accept: asymmetric encryption is used to exchange a symmetric session key securely (as in TLS).

4 Describe how a digital signature is created and verified. [4]

> The sender creates a hash (digest) of the message (1) / and encrypts the hash with the sender's private key – this is the signature, sent with the message (1) / the receiver decrypts the signature with the sender's public key to recover the hash (1) / and hashes the received message themselves; if the two hashes match, the message is authentic (from the sender) and has not been altered (1).

5 State the purpose of a digital certificate. [2]

> A digital certificate is issued by a trusted certificate authority (CA) (1) / to bind a public key to the identity of its owner (e.g. a bank's domain), so others can trust that the public key genuinely belongs to that organisation (1).

6 A firm's staff work from home. Evaluate the use of a VPN and multi-factor authentication to protect the firm's network. [6]

> Indicative content. VPN: creates an encrypted tunnel across the internet so data cannot be read if intercepted, e.g. on home or café Wi-Fi / lets staff use internal systems as if on site; access can be controlled centrally / drawbacks: adds latency, needs configuration and support, depends on trust in the VPN provider, and does not protect a device already infected with malware. MFA: requires two or more factors (know, have, are), e.g. password plus code on phone, so a stolen or phished password alone is not enough / greatly reduces

unauthorised access / drawbacks: slight inconvenience, cost of tokens/apps, risk of lockout if a phone is lost, users may be targeted by MFA-fatigue attacks. Together: MFA protects the login, VPN protects data in transit – complementary. Neither stops malware, insider misuse or poor practice, so also need firewalls, anti-malware, patching and training. Conclusion: both are strongly recommended as part of layered security. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 Alice wants to send Bob a secret message. Which key does she encrypt it with? [1]

A Alice's private key
B Bob's public key ✓
C Bob's private key
D Alice's public key

2 Who issues digital certificates? [1]

A Internet service providers
B Web browsers
C Certificate authorities ✓
D Firewalls

3 Flooding a server with traffic from a botnet is... [1]

A phishing
B a DDoS attack ✓
C a man-in-the-middle attack
D spyware

4 A fingerprint scan is authentication by... [1]

A something you know
B something you have
C something you are ✓
D something you share

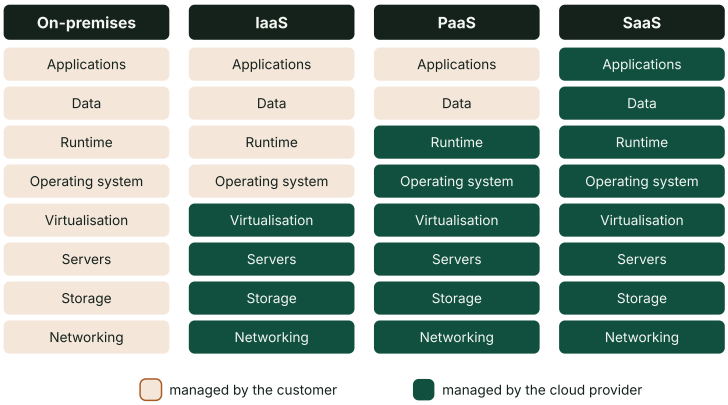
5 Which is NOT normally a function of a proxy server? [1]

A Caching pages
B Hiding internal IP addresses
C Filtering content
D Issuing digital certificates ✓

Cloud computing and virtualisation

IS14 ANSWERS

I can: Distinguish IaaS, PaaS and SaaS by what the provider and customer each manage · Recommend a public, private or hybrid cloud for a scenario with justification · Explain server virtualisation and evaluate its benefits and drawbacks



Use this diagram to help you: Who manages each layer of the computing stack in on-premises, IaaS, PaaS and SaaS models: the customer or the cloud provider

A Key terms

Write the letter of the matching definition · 6 marks

A Cloud computing

B Infrastructure as a service (IaaS)

C Platform as a service (PaaS)

E Software as a service (SaaS)

F Hybrid cloud

D Virtualisation

A The on-demand delivery of computing resources such as servers, storage and software over the internet, usually paid for as used.

B A cloud model in which the provider supplies virtualised hardware such as servers, storage and networking, and the customer manages the operating system and software.

C A cloud model in which the provider supplies the hardware, operating system and development platform so customers only build and manage their own applications.

D The use of software to create virtual versions of hardware, allowing several virtual machines to run on one physical computer.

E A cloud model in which complete application software is hosted by the provider and accessed by users over the internet.

F A cloud deployment that combines a private cloud with public cloud services, allowing data and applications to move between them.

B Fill the gaps

Use the word bank · 5 marks

hypervisor | SaaS | scalability | private | service

Renting virtual servers and storage from a provider is infrastructure as a **service**. Microsoft 365 is an example of **SaaS**. A cloud used by only one organisation is a **private** cloud. Software

that runs several virtual machines on one physical server is a **hypervisor** . The ability to add capacity quickly as demand grows is **scalability** .

C True or false?

Tick one box · 5 marks

	True	False
In SaaS the customer manages the operating system. *(In SaaS the provider manages everything except the customer's data and settings.)*	☐	☑
A hybrid cloud combines private and public clouds.	☑	☐
Virtualisation allows several operating systems to run on one physical server.	☑	☐
Using a cloud provider means an organisation no longer has any responsibilities under data protection law. *(The organisation remains the controller and must ensure the provider processes data lawfully and securely.)*	☐	☑
Cloud computing depends on a reliable internet connection.	☑	☐

D Exam-style questions

19 marks

1 Describe what is meant by software as a service (SaaS) and give an example. [2]

> SaaS: the complete application is provided over the internet by the provider, who manages the software, updates and infrastructure; the customer manages only their data/settings (1) / example: Microsoft 365, Gmail, Google Classroom (1).

2 Explain the difference between IaaS and PaaS. [2]

> IaaS provides virtualised infrastructure – servers, storage and networking – and the customer installs and manages the operating system, runtime and applications, e.g. renting virtual machines (1) / PaaS provides a platform including the OS, runtime and database, so the customer only develops and manages their own applications and data, e.g. deploying a web app to Azure App Service (1).

3 Describe a hybrid cloud and give one reason an organisation might choose it. [2]

> A hybrid cloud combines private cloud (or on-site) infrastructure with public cloud, with data and applications moving between them (1) / reason, any one: keep sensitive data in the private cloud while using the public cloud for less sensitive work or for peak demand (cloud bursting) (1).

4 Explain what is meant by virtualisation and the role of a hypervisor. [3]

> Virtualisation uses software to create virtual versions of hardware, such as servers, storage or networks (1) / so one physical machine can run several virtual machines, each with its own operating system (1) / the hypervisor is the software that creates, runs and manages the VMs, allocating the physical resources (CPU, memory, storage) between them and keeping them isolated (1). Accept type 1 (bare-metal) / type 2 (hosted) distinction.

5 Give two benefits of server virtualisation to an organisation. [2]

> Any two from: server consolidation – fewer physical servers, so lower costs for hardware, power, cooling and space (1) / new servers can be provisioned in minutes (1) / VMs are isolated, so a crash in one does not affect others and testing is safe (1) / snapshots and migration make backup and disaster recovery easier (1).

6 Discuss the advantages and disadvantages for a GP surgery of storing patient records in a public cloud. [8]

> Indicative content. Advantages: provider manages hardware, updates, backups and physical security, so less need for in-house IT staff / lower upfront cost, pay-as-you-go / records accessible from anywhere (home visits, other surgeries, out-of-hours) / scalable storage / disaster recovery across data centres, high availability / professional security often better than a small surgery could provide. Disadvantages: patient records are special

category (health) data under UK GDPR, so the surgery (controller) remains responsible and must ensure the provider (processor) has a contract and strong security / data sovereignty – data may be stored outside the UK under different laws / depends on a reliable internet connection; outage could stop access to records during appointments (an issue in rural Fermanagh) / risk of breach or unauthorised access; shared infrastructure / vendor lock-in and ongoing subscription costs / less control over changes and downtime. Possible mitigations: encryption, MFA, UK-based data centres, a hybrid or private cloud for the most sensitive data. Conclusion: justified judgement. Level 3 (6–8): thorough, balanced discussion of both sides, clearly applied to the context, with a justified conclusion; QWC: fluent, well structured, accurate specialist terms. Level 2 (3–5): a range of relevant points with some explanation and application; may be unbalanced or lack a conclusion; QWC: generally clear. Level 1 (1–2): limited, mostly descriptive points with little application; QWC: basic, errors may obscure meaning.

E Quick quiz

Circle one answer · 5 marks

1 A developer deploys web app code without managing any servers or operating systems. This is... [1]

A IaaS
B PaaS ✓
C SaaS
D on-premises

2 Which deployment model is shared by many organisations? [1]

A Public cloud ✓
B Private cloud
C Hybrid cloud
D Intranet

3 Which is a drawback of cloud computing? [1]

A Scalability
B Access from anywhere
C Dependence on internet connectivity ✓
D No upfront hardware cost

4 What does a hypervisor do? [1]

A Encrypts cloud data
B Creates and runs virtual machines ✓
C Assigns IP addresses
D Filters web traffic

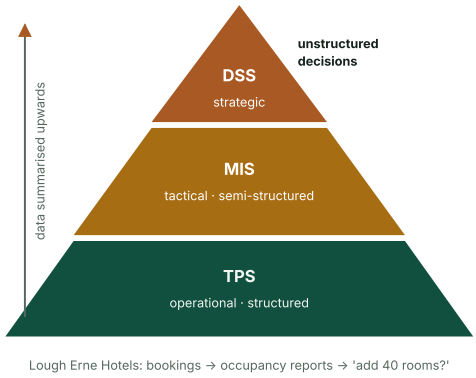
5 Data stored in another country being subject to that country's laws is an issue of... [1]

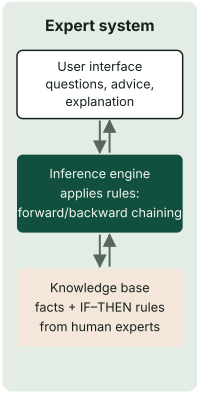
A elasticity
B virtualisation
C redundancy
D data sovereignty ✓

Types of information system

IS15 ANSWERS

I can: Match types of information system to the levels of management they serve · Explain how a TPS, MIS and DSS work together using an example organisation · Describe the components of an expert system and evaluate its use





Use this diagram to help you: The information pyramid: TPS at the operational level, MIS at the tactical level and DSS at the strategic level, with data flowing upwards; and the three components of an expert system

A Key terms

Write the letter of the matching definition · 6 marks

A Transaction processing system (TPS)

D Management information system (MIS)

B Decision support system (DSS)

C Expert system

F Knowledge base

E Inference engine

A An information system that records and processes the routine day-to-day transactions of an organisation, in real time or in batches.

B An interactive information system that uses data and models to help managers make semi-structured and unstructured decisions, for example through what-if analysis.

C A computer system that emulates the decision-making of a human expert in a specific domain using a knowledge base and an inference engine.

D An information system that summarises data from transaction systems into regular, exception and on-demand reports for middle managers.

E The part of an expert system that applies the rules in the knowledge base to the facts supplied in order to reach conclusions.

F The part of an expert system that stores facts and rules gathered from human experts.

B Fill the gaps

Use the word bank · 5 marks

knowledge | inference | decision | batch | transaction

A **transaction** processing system records day-to-day sales and bookings. Processing all payroll records together overnight is **batch** processing. A **decision** support system allows

managers to carry out what-if analysis. In an expert system, the **inference** engine applies rules to facts. The facts and rules are stored in the **knowledge** base.

C True or false?

Tick one box · 5 marks

	True	False
A TPS is mainly used by directors to make strategic decisions. *(A TPS supports operational staff with routine transactions; strategic decisions use DSS/EIS.)*	☐	☑
An exception report highlights results outside normal limits.	☑	☐
Goal seeking works backwards from a desired result to find the input needed.	☑	☐
An expert system can use common sense in situations outside its rules. *(It can only apply the rules in its knowledge base.)*	☐	☑
Forward chaining starts from the known facts and works towards a conclusion.	☑	☐

D Exam-style questions

21 marks

1 State the difference between batch and real-time transaction processing, giving an example of each. [4]

> Batch processing: transactions are collected together and processed as a group later, e.g. at night (1) / example: payroll or utility billing (1) / Real-time processing: each transaction is processed immediately and records are updated straight away (1) / example: cinema seat booking or bank withdrawal at an ATM (1).

2 Describe three types of report that a management information system might produce. [3]

> Scheduled (periodic) reports produced at regular intervals, e.g. weekly sales by branch (1) / exception reports produced only when something falls outside normal limits, e.g. stores below target or hotel occupancy under 60% (1) / on-demand (ad hoc) reports produced when a manager requests them, e.g. a query on a specific product (1). Accept summary/detailed reports with example.

3 Explain how a decision support system differs from a management information system. [3]

> An MIS produces regular, structured reports, mostly from internal historical data, to support structured/semi-structured tactical decisions (1) / a DSS supports semi-structured or unstructured (often strategic) decisions using internal and external data and a model base (1) / it is interactive, allowing what-if analysis, goal seeking and forecasting, e.g. 'what happens to profit if fuel prices rise 10%?' (1).

4 Describe the three main components of an expert system. [3]

> Knowledge base: stores the facts and IF...THEN rules gathered from human experts (1) / Inference engine: applies the rules to the facts entered by the user to reach conclusions, using forward or backward chaining (1) / User interface: asks the user questions and presents conclusions, often with an explanation facility showing the reasoning (1).

5 Discuss the use of an expert system to help GPs diagnose illnesses. [8]

> Indicative content. How it works: knowledge base of symptoms, test results and diagnostic rules from consultants; inference engine applies rules to the patient's symptoms entered by the GP; user interface asks questions and gives likely diagnoses with an explanation facility. Benefits: expertise available 24/7 and in many surgeries at once, including rural ones / consistent, does not tire or forget, may suggest rare conditions a GP might miss / explanation facility supports training of junior doctors / can preserve knowledge of retiring specialists / may speed diagnosis and reduce referrals. Drawbacks: lacks common sense and cannot handle cases outside its rules; cannot see the patient (body language, examination) / expensive and slow to build; knowledge hard to capture from experts / must be kept up to date as medical knowledge changes / responsibility unclear if advice is wrong / GPs may over-trust it or deskill / patients may lose trust; health data is special category data under UK GDPR and must be secured. Conclusion: valuable as a decision-support tool,

with the GP keeping final responsibility (human in the loop). Level 3 (6–8): thorough, balanced discussion of both sides, clearly applied to the context, with a justified conclusion; QWC: fluent, well structured, accurate specialist terms. Level 2 (3–5): a range of relevant points with some explanation and application; may be unbalanced or lack a conclusion; QWC: generally clear. Level 1 (1–2): limited, mostly descriptive points with little application; QWC: basic, errors may obscure meaning.

E Quick quiz

Circle one answer · 5 marks

1 A cinema seat booking updated immediately is an example of... [1]

A batch processing
B real-time transaction processing ✓
C a DSS
D an expert system

2 Which system produces weekly sales reports for each branch? [1]

A MIS ✓
B TPS
C Expert system
D Hypervisor

3 'What would profit be if we cut prices by 5%?' uses... [1]

A an exception report
B forward chaining
C the knowledge base
D what-if analysis ✓

4 Which part of an expert system shows how a conclusion was reached? [1]

A Knowledge base
B Explanation facility ✓
C Transaction log
D Model base

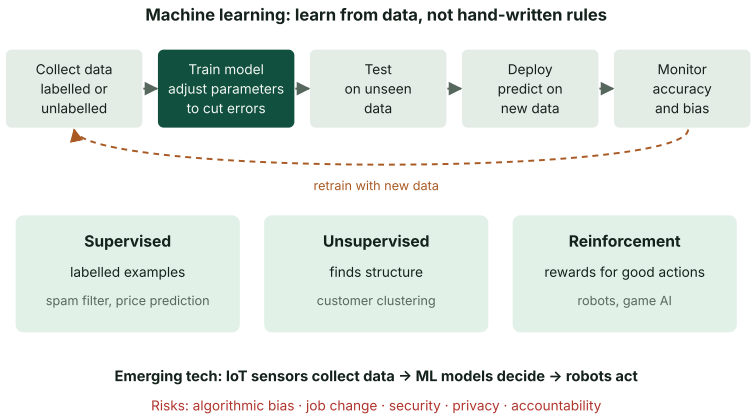
5 Who builds the knowledge base by interviewing experts? [1]

A Database administrator
B Network manager
C Knowledge engineer ✓
D Data subject

Artificial intelligence, machine learning and emerging technologies

IS16 ANSWERS

I can: Explain what artificial intelligence and machine learning are and how a model is trained · Describe the Internet of Things and robotics with organisational examples · Evaluate the benefits and risks of AI and emerging technologies for organisations and employees



Use this diagram to help you: The machine learning cycle of training data, training, testing on unseen data, deployment and monitoring, with the three types of learning

A Key terms

Write the letter of the matching definition · 5 marks

B Artificial intelligence (AI)

A Machine learning

C Supervised learning

E Algorithmic bias

D Internet of Things (IoT)

A A branch of AI in which algorithms improve their performance at a task by learning patterns from data rather than following explicitly programmed rules.

B The ability of computer systems to perform tasks that normally require human intelligence, such as recognising images, understanding language and making decisions.

C Machine learning in which a model is trained on labelled examples whose correct outputs are known.

D A network of everyday physical objects with embedded sensors, processors and connectivity that collect and exchange data.

E Systematic unfairness in the outputs of an AI system, usually caused by biased or unrepresentative training data.

B Fill the gaps

Use the word bank · 5 marks

reinforcement | Things | supervised | bias | Machine

Machine learning allows a system to learn from data rather than being explicitly programmed. Learning from labelled examples is **supervised** learning. Learning through rewards and trial and error is **reinforcement** learning. Unfair results caused by unrepresentative training data are called algorithmic

bias . Everyday objects connected to the internet with sensors form the Internet of **Things** .

C True or false?

Tick one box · 5 marks

	True	False
All current AI systems are examples of narrow AI.	☑	☐
Unsupervised learning needs every training example to be labelled with the correct answer. *(That describes supervised learning; unsupervised learning uses unlabelled data.)*	☐	☑
A model should be tested on data it was not trained on.	☑	☐
IoT devices are always secure because they are small. *(Many IoT devices have weak security and can be recruited into botnets.)*	☐	☑
Reinforcement learning uses rewards to improve behaviour.	☑	☐

D Exam-style questions

21 marks

1 Define machine learning. [2]

> Machine learning is a type of AI in which algorithms improve their performance at a task by learning from data (1) / rather than being explicitly programmed with rules – the model finds patterns in training data and uses them to make predictions on new data (1).

2 Distinguish between supervised and unsupervised learning, giving an example of each. [4]

> Supervised learning trains the model on labelled data, i.e. inputs with the correct outputs (1) / example: emails labelled spam/not spam to classify new emails; predicting house prices (1) / Unsupervised learning finds patterns or structure in unlabelled data with no correct answers given (1) / example: clustering customers into segments for marketing (1).

3 Explain what is meant by algorithmic bias and how it can occur. [2]

> Algorithmic bias is when an AI system makes unfair or discriminatory decisions that systematically favour or disadvantage particular groups (1) / it can occur when the model is trained on biased or unrepresentative data, e.g. a recruitment model trained on past hiring that favoured men (1). Accept: biased choice of features or design by developers.

4 Describe the Internet of Things and give one example of its use by an organisation. [3]

> The IoT is a network of everyday physical objects fitted with sensors, processors and network connections (1) / which collect and exchange data with other devices or systems over the internet without human input (1) / example, any one: sensors tracking cattle on Fermanagh farms / smart meters for energy firms / smart heating in schools / sensors on machinery for predictive maintenance / tracking stock or vehicles in logistics (1).

5 Evaluate the impact of introducing robots and AI in a large warehouse on the business and its employees. [10]

> Indicative content. Business – positives: automation of picking and packing increases productivity and allows 24/7 operation / fewer errors, faster order fulfilment and better customer service / AI forecasting improves stock control; IoT sensors support predictive maintenance / lower long-term labour costs; competitive advantage. Business – negatives: high set-up cost and specialist staff / dependence on suppliers, data quality and reliable systems; breakdowns or cyber attacks can halt operations / IoT devices may have weak security / reputational risk if seen to cut jobs. Employees – positives: robots take over heavy, repetitive and dangerous work, improving health and safety / new, higher-skilled jobs in maintenance, programming and data / chance to retrain and upskill. Employees – negatives: job losses in routine roles / deskilling of remaining work / need to retrain, which older staff may find hard / monitoring of productivity by AI may cause stress and privacy concerns (UK GDPR) / algorithmic decisions about shifts or performance may be biased or opaque. Judgement: e.g. benefits to the

business are large, but success depends on retraining and fair treatment of staff. Level 3 (8–10): thorough evaluation of impacts on both the business and its employees, positive and negative, well applied to the warehouse, with a justified overall judgement; QWC: fluent, logically structured, specialist terms used accurately. Level 2 (4–7): a range of relevant impacts explained with some application; may focus mainly on one stakeholder or lack a judgement; QWC: generally clear with some specialist terms. Level 1 (1–3): a few simple points, largely descriptive with little application; QWC: basic, with errors that may obscure meaning.

E Quick quiz

Circle one answer · 5 marks

1 Grouping customers into segments without predefined labels is... [1]

A supervised learning
B reinforcement learning
C unsupervised learning ✓
D an expert system

2 Which is an IoT example? [1]

A A spreadsheet
B A smart meter sending readings to the supplier ✓
C A printed report
D A USB stick

3 A recruitment AI rejects more women because past hiring data favoured men. This is... [1]

A overfitting
B data redundancy
C encryption
D algorithmic bias ✓

4 What happens during training? [1]

A The model adjusts its parameters to reduce errors on training data ✓
B The model is deleted
C Data is normalised to 3NF
D Users enter rules by hand

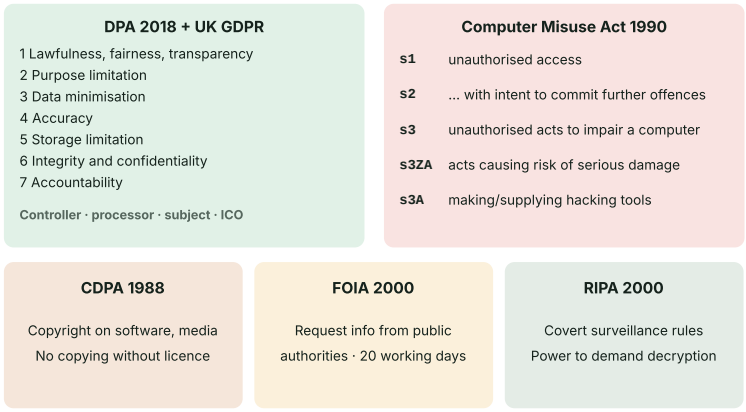
5 Which is a likely impact of automation on employment? [1]

A No jobs change
B Routine jobs decline and new skilled jobs appear ✓
C All jobs disappear
D Wages always rise

Legislation: data protection, computer misuse, copyright, FOI and RIPA

IS17 ANSWERS

I can: State and apply the seven data protection principles and the rights of data subjects · Identify which offence under the Computer Misuse Act applies to a scenario · Explain how copyright, freedom of information and investigatory powers legislation affect organisations and individuals



Use this diagram to help you: Summary of the five Acts: Data Protection Act 2018 and UK GDPR, Computer Misuse Act 1990, Copyright Designs and Patents Act 1988, Freedom of Information Act 2000 and RIPA 2000

A Key terms

Write the letter of the matching definition · 6 marks

E Personal data	**A** UK law creating offences of unauthorised access to computer material, access with intent to commit further offences, and unauthorised acts that impair computers.
C Data controller	**B** The living individual whom personal data is about.
D Data processor	**C** The person or organisation that decides the purposes for which, and the way in which, personal data is processed.
B Data subject	**D** A person or organisation that processes personal data on behalf of a data controller.
F Special category data	**E** Any information relating to an identified or identifiable living individual.
A Computer Misuse Act 1990	**F** Personal data needing extra protection, such as health, ethnicity, religion, political opinions, trade union membership, genetic and biometric data.

B Fill the gaps

Use the word bank · 5 marks

unauthorised · minimisation · public · ICO · controller

The organisation that decides how personal data is processed is the data **controller**. Collecting only the data needed follows the principle of data **minimisation**. In the UK, data protection is regulated by the **ICO**. Section 1 of the Computer Misuse Act covers **unauthorised** access to computer material. The Freedom of Information Act applies to **public** authorities.

C True or false?

Tick one box · 6 marks

	True	False
Health data is special category data under UK GDPR.	☑	☐
The Freedom of Information Act lets anyone request information held by private companies. *(It applies to public authorities, not private companies.)*	☐	☑
Spreading ransomware is an offence under section 3 of the Computer Misuse Act.	☑	☐
Copyright must be registered before a program is protected. *(Copyright arises automatically when the work is created.)*	☐	☑
Data protection law applies to data about deceased people. *(UK GDPR covers living individuals only.)*	☐	☑
RIPA includes a power to require encrypted data to be decrypted.	☑	☐

D Exam-style questions

18 marks

1 State four of the seven data protection principles. [4]

> Any four from: lawfulness, fairness and transparency (1) / purpose limitation (1) / data minimisation (1) / accuracy (1) / storage limitation (1) / integrity and confidentiality (security) (1) / accountability (1).

2 Explain the difference between a data controller and a data processor. [2]

> A data controller decides why and how personal data is processed (the purposes and means) and is responsible for compliance, e.g. a school (1) / a data processor processes personal data on behalf of the controller, following its instructions, e.g. a cloud provider or payroll company (1).

3 A pupil uses a teacher's password to log in and change their exam marks. Identify which offences under the Computer Misuse Act 1990 have been committed and explain why. [3]

> Section 1: unauthorised access to computer material – the pupil logged in with a password they were not entitled to use (1) / Section 2: unauthorised access with intent to commit or facilitate a further offence – they accessed the system intending to change their marks, i.e. fraud/deception (1) / Section 3: unauthorised acts with intent to impair the operation of a computer / the reliability of data – they altered the exam marks stored on the system (1).

4 Describe two rights of a data subject. [4]

> Any two rights, each named (1) and described (1): right of access (1) / to request a copy of the personal data held about them via a subject access request, normally answered within one month (1) / right to rectification (1) / to have inaccurate or incomplete data corrected (1) / right to erasure ('right to be forgotten') (1) / to have data deleted in certain circumstances, e.g. no longer needed (1) / right to object (1) / e.g. to stop processing for direct marketing (1) / right to be informed (1) / to be told how and why their data is used, e.g. a privacy notice (1) / right to data portability (1) / to obtain and reuse their data with another provider (1) / right to restrict processing (1) / rights related to automated decision-making and profiling (1).

5 Explain the purpose of the Freedom of Information Act 2000. [2]

> It gives anyone the right to request recorded information held by public authorities, e.g. councils, schools, health trusts (1) / to promote openness, transparency and accountability of public bodies; the authority must normally respond within 20 working days, subject to exemptions such as personal data or national security (1).

6 A software company discovers that a school has installed one licensed copy of its program on 50 computers. Explain which law has been broken and the possible consequences. [3]

> The Copyright, Designs and Patents Act 1988 has been broken (1) / because the school has copied (installed) the software 49 more times than its single-user licence allows, without the copyright owner's permission (1) /

consequences, any one: the company could take legal action for damages / the school may have to pay for 49 more licences or a site licence / fines, and in serious cases criminal prosecution / reputational damage (1).

E Quick quiz

Circle one answer · 5 marks

1 Keeping applicants' CVs for ten years with no reason breaches... [1]

A accuracy
B storage limitation ✓
C data portability
D purpose limitation only

2 Writing and selling malware kits is an offence under... [1]

A CMA section 3A ✓
B CDPA 1988
C FOIA 2000
D CMA section 1

3 A cloud provider storing a school's data on its behalf is a... [1]

A data subject
B data controller
C data processor ✓
D Information Commission

4 A public authority should normally respond to an FOI request within... [1]

A 5 working days
B 40 calendar days
C 6 months
D 20 working days ✓

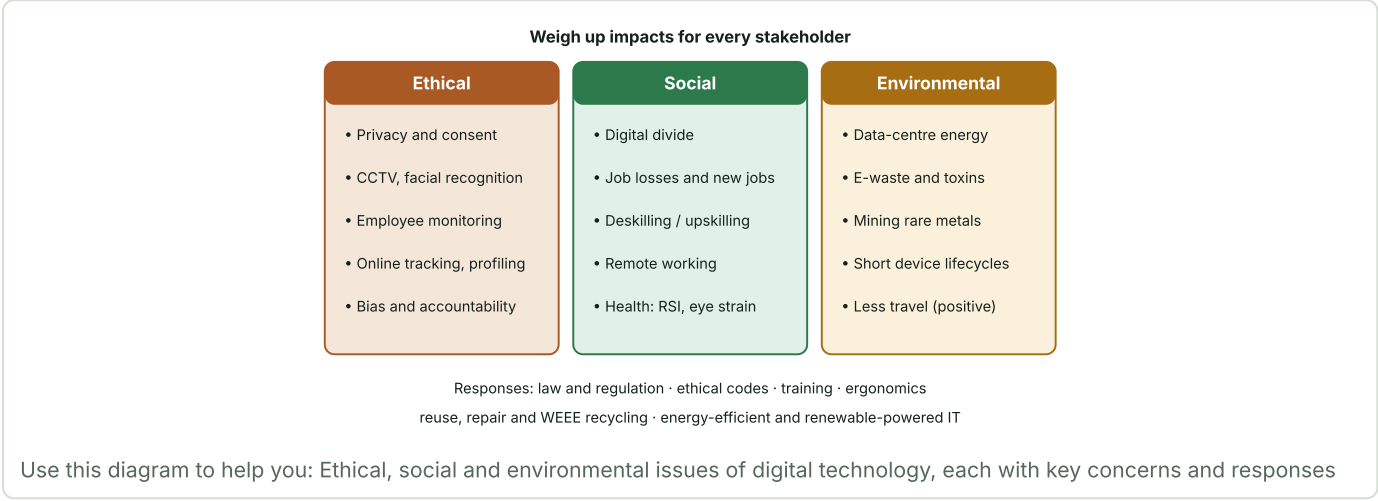
5 Which Act protects software from being copied without permission? [1]

A Computer Misuse Act 1990
B Copyright, Designs and Patents Act 1988 ✓
C Freedom of Information Act 2000
D RIPA 2000

Ethical, social and environmental issues

IS18 ANSWERS

I can: Explain privacy and surveillance issues from the viewpoints of different stakeholders · Discuss the causes and effects of the digital divide and the impact of technology on employment and health · Evaluate the environmental impact of digital technology and ways of reducing it



Use this diagram to help you: Ethical, social and environmental issues of digital technology, each with key concerns and responses

A Key terms

Write the letter of the matching definition · 5 marks

D Digital divide

A E-waste

B Surveillance

E Repetitive strain injury (RSI)

C Sustainability

A Discarded electrical and electronic devices, which may contain toxic and valuable materials.

B The monitoring of people's behaviour, activities or communications, for example by CCTV, tracking or interception.

C Meeting present needs without compromising the ability of future generations to meet theirs, for example by reducing energy use and e-waste.

D The gap between people who have access to digital technology and the skills to use it, and those who do not.

E Pain or damage to muscles, tendons or nerves caused by repeated movements such as typing or using a mouse.

B Fill the gaps

Use the word bank · 5 marks

e-waste | privacy | injury | deskilling | divide

The gap between those with and without access to technology is the digital **divide**. Discarded electronic devices are known as **e-waste**. Repeated use of a keyboard can cause repetitive strain **injury**. Monitoring people using CCTV and tracking raises concerns about **privacy**. When technology does the skilled part of a job, workers may experience **deskilling**.

C True or false?

Tick one box · 5 marks

	True	False

Something that is legal is always ethical. *(The law sets minimum standards; some legal practices are still considered unethical.)*	☐	☑
E-waste can contain toxic materials such as lead and mercury.	☑	☐
Video conferencing can reduce an organisation's carbon footprint by cutting travel.	☑	☐
The digital divide only exists between countries. *(It also exists within countries, e.g. between urban and rural areas or age groups.)*	☐	☑
Regular breaks help reduce eye strain from screen use.	☑	☐

D Exam-style questions

29 marks

1 Define the term digital divide. [2]

> The digital divide is the gap between people who have access to digital technology (devices and internet) (1) / and the skills/confidence to use it, and those who do not (1).

2 Give two causes of the digital divide in Northern Ireland and suggest a way to reduce each. [4]

> Two causes (1 each) with a matching solution (1 each), e.g.: poor rural broadband connectivity, e.g. in parts of Fermanagh (1) / government broadband roll-out schemes (e.g. Project Stratum) or satellite/4G-5G alternatives (1) / cost of devices and broadband for low-income households (1) / device lending schemes, subsidised social tariffs (1) / lack of skills or confidence, especially among older people (1) / free digital-skills classes in libraries and community centres (1) / disability (1) / accessible design and assistive technology (1).

3 Describe two health problems that can result from prolonged computer use and how each can be prevented. [4]

> Two problems (1 each) with prevention (1 each), e.g.: repetitive strain injury (RSI) from repeated keyboard/mouse use (1) / ergonomic keyboards, wrist rests and regular breaks (1) / eye strain and headaches from long periods at screens and glare (1) / anti-glare screens, correct lighting, breaks, free eye tests (1) / back and neck problems from poor posture or unsuitable chairs (1) / adjustable chairs and screens, posture training (1) / stress (1) / workload management, right-to-disconnect policies (1). Accept reference to the Health and Safety at Work (NI) Order 1978 / display screen equipment regulations.

4 Explain two ways an organisation can reduce the environmental impact of its IT. [4]

> Two ways (1 each) explained (1 each), e.g.: extend device lifecycles by repairing, reusing and refurbishing (1) / reducing manufacturing demand for energy and rare earth metals and reducing e-waste (1) / recycle old equipment through approved schemes under the WEEE Regulations (1) / so toxic materials such as lead and mercury do not go to landfill (1) / use energy-efficient hardware and power-management settings / virtualise servers (1) / reducing electricity used for processing and cooling (1) / choose cloud providers using renewable energy (1) / reducing the carbon footprint of data storage (1) / use video conferencing instead of travel (1) / reducing emissions from commuting and business travel (1).

5 Discuss the ethical, social and environmental issues raised by an employer monitoring staff who work from home. [15]

> Indicative content. Ethical: right to privacy vs employer's right to protect its business / proportionality and transparency – staff must be told what is monitored and why / covert monitoring (keystrokes, webcams, screenshots) is intrusive and erodes trust / home monitoring may capture family members and private life / under UK GDPR monitoring must have a lawful basis, be necessary and proportionate, and a DPIA is usually needed; employees have rights of access and to object; the Information Commission (formerly the ICO) regulates / automated performance judgements may be biased or unfair / who has access to the data and how long it is kept (storage limitation). Social: protects company data and detects security breaches; can show fair workload and productivity / but causes stress, anxiety and an 'always on' culture; blurs work–home boundaries; affects mental health / may damage morale and trust, leading to higher staff turnover / remote work benefits (no commute, flexibility, wider talent pool, inclusion of carers and disabled workers) could be undermined / employees may feel isolated. Environmental: home working reduces commuting, fuel use and emissions, and office energy use / but monitoring software and storage of large volumes of logs/video increase energy use in

> data centres / more devices in homes and duplicate equipment increase e-waste. Conclusion: monitoring can be justified if limited, transparent, proportionate and lawful, with focus on outcomes rather than surveillance. Level 3 (11–15): thorough discussion covering all three areas (ethical, social and environmental), balanced between employer and employee viewpoints, well applied to home working and supported by examples, with a reasoned conclusion; QWC: fluent, well structured, specialist terms (e.g. UK GDPR, proportionality, DPIA) used accurately. Level 2 (6–10): relevant points across at least two areas with some explanation and application; discussion may be unbalanced or the conclusion weak; QWC: generally clear and organised. Level 1 (1–5): a few simple points, mostly descriptive, perhaps from one area only; QWC: basic, with errors that may obscure meaning.

E Quick quiz

Circle one answer · 5 marks

1 Which is a cause of the digital divide? [1]

A Fast fibre broadband in every home
B Poor rural connectivity ✓
C Cheap devices for all
D Universal digital skills

2 Recycling electrical equipment in the UK is governed by the... [1]

A WEEE Regulations ✓
B Computer Misuse Act
C FOI Act
D RIPA

3 Which health problem is linked to repeated mouse use? [1]

A Eye strain
B Hearing loss
C Repetitive strain injury ✓
D Deskilling

4 A self-checkout replacing cashiers is an example of... [1]

A upskilling
B digital divide
C e-waste
D automation affecting employment ✓

5 Which reduces the environmental impact of IT? [1]

A Replacing laptops every year
B Server virtualisation ✓
C Leaving PCs on overnight
D Sending old devices to landfill